WDI
WISE DEVICE INC

ATF 6.5 Sensor User Guide



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WDI Wise Device Inc.
135 West Beaver Creek Road • Suite 2
Richmond Hill, ON L4B 1L2
Phone 905 415 2734 • Fax 905 415 5973
www.wdidevice.com

Technical support: *applications@wdidevice.com*

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List of acronyms

ATF	Autofocus
AOI	Automated optical inspection
AR	Anti-reflective coating
CCW	Counter-clockwise
CF	Color filter
CM	Cell and module
CR	Capture range
CW	Clockwise
DIO	Digital input and output
DOF	Depth of field
ESD	Electrostatic discharge
FCC	Federal Communication Commission
FOV	Field of view
FPGA	Field-programmable gate array
LED	Light emitting diode
LLC	Linear lens changer
LR	Linear range
MCM	Motor control module
MEMS	Microelectromechanical system
NA	Numerical aperture
ND	Neutral density
OD	Optical density
P	Proportional gain constant
PWM	Pulse width modulation
SNR	Signal-to-noise ratio
TFT	Thin film transistor
T/R	Transmission/reflection ratio
VFC	Video focus configuration
ZAA	Z-axis actuator

CHAPTER

1

Introduction

This ATF 6.5 sensor is a digital autofocus sensor that provide fast and accurate measurement of the distance and direction to focus. This chapter provides an overview of the sensor, as well as some general product and company information.

The following topics are covered:

- *Sensor description, pg. 10*
  - *Principle of operation, pg. 10*
  - *Improvements since ATF 6, pg. 11*
  - *ATF 6.5 and ATF 6 comparison, pg. 11*
- *Intended use, pg. 13*
  - *Laser classification, pg. 13*
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- *Changes to WDI products, pg. 15*
- *Warranty policy, pg. 15*
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  - *Warranty repair process, pg. 15*
  - *Disclaimer, pg. 16*
- *Technical support, pg. 16*
- *Safety notices, pg. 16*

Sensor description

The ATF 6.5 sensor is an automated tracking focus sensor that allows focusing servo systems with fast feedback signals to quickly and accurately focus almost any type of infinity-corrected microscope. The sensor determines the direction towards focus and measures the distance to the best focus position. This information is sent to a controller, which uses the data to move a motorized Z-stage, positioning the sample at the best focus position. Combining the sensor with a controller and motorized stage in this way creates a complete autofocus system.

Since the ATF 6.5 sensor measures the distance to focus, it is effective regardless of whether the specimen is stationary or in motion. This feature, called tracking autofocus, is particularly useful when multiple images of spatially distributed areas must be quickly collected from a specimen. In such cases, tracking autofocus eliminates the focusing overhead time and improves the instrument throughput.

Principle of operation

The ATF 6.5 sensor uses the principle of through-the-lens (TTL) triangulation to assess the direction and distance to focus. A laser beam is projected through the objective lens of the microscope. After passing through the objective, the laser beam is reflected by the sample. The resulting image is analyzed by the sensor and converted by the software to a distance and direction to focus measurement. When the laser beam appears as a thin line, the specimen is in the best focus position. When the laser beam expands into more of a rectangle, either to the left or right of the main optical axis, the specimen is above or below the best focus position, respectively.

By projecting a reference laser beam through the microscope's own objective lens, the sensor eliminates the need for surface features to judge the degree of focusing. Thus the sensor can operate effectively on both patterned and unpatterned specimens, and can tolerate diffuse reflective surfaces, including thick and thin glass sheets. For three-dimensional periodical structures, such as MEMS or touch screens, you can choose to keep focus on either the substrate or high spots.

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Improvements since ATF 6

The ATF 6.5 sensor includes significant improvements over the ATF 6 sensor in hardware, software, and functionality, with additional features such as advanced surface recognition, multi-plane autofocus, and video autofocus.

The increased laser power of the ATF 6.5 sensor (up to 2.5 mW) has the following benefits:

- Permits the sensor to achieve a larger Capture Range (CR) for Cell and Module (CM) substrates, particularly with 20X and 50X objective lenses.
- Provides a wider beam, which makes the sensor more forgiving with respect to microscope angular accuracy, resulting in easier alignment.
- Permits integration with a microscope coupling filter that does not require kinematic adjustments. This facilitates integration at further distances to the back aperture of the objective lens—up to 300 mm compared to 200 mm for the ATF 6.
- Allows for integration with a 50/50 beamsplitter rather than a 30/70 beamsplitter, which is required for the ATF 6.
- Can overcome bright field illumination and tolerate higher levels of ambient light, which improves and speeds up video autofocus functionality.

An additional benefit of the upgraded ATF 6.5 sensor optics is minimized astigmatism distortion, which can affect the CR and CM at high magnification lenses, particularly when the sensor is aligned further away from the objective lens.

ATF 6.5 and ATF 6 comparison

Table 1 provides a comparison of ATF 6.5 and ATF 6 features.

Table 2 compares the optic-specific performance data of the ATF 6.5 and ATF 6 sensor. The data were gathered using long working distance Mitutoyo M Plan APO 5X, 10X, 20X, and 50X objective lenses.

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Table 1 ATF 6.5 and ATF 6 feature comparison

Features	ATF 6.5 sensor	ATF 6 sensor
Optical		
Laser wavelength	680 nm or 785 nm	660 nm or 785 nm
Power output	2.5 mW maximum	0.25 mW maximum
Sensor coupling		
Notch filter	YES	YES
50/50 beamsplitter	YES	NO
Maximum standoff distance to objective lens back focal plane	300 mm	200 mm
Modes		
Surface recognition	Standard, up to 50X	Add ON, only on 5X and 20X
SWIFT	Standard	Add ON
Multi-plane autofocus (MPAF)	Standard	Standard
PZ	Add ON	Add ON
2X objective lens	Direct switching	Step switching[a]

a. *When using a 50X objective lens, you must step down to a lower magnification lens (e.g., 5X or 10X) before switching to a 2X lens.*

Table 2 ATF 6.5 and ATF 6 performance data

Objective Lens	Numerical Aperture	DOF [µm][a]	Linear Range[b] [µm]		Capture Range[c] [µm]	
			ATF6.5	ATF6	ATF6.5	ATF6
5X	0.14	±14	±420	±560	±5000	±3000
10X	0.28	±3.5	±100	±130	±3000	±1500
20X	0.42	±1.6	±25	±30	±1800	±600
50X	0.55	±0.9	±10	±10	±750	±250

a. *Focus repeatability is normally 1/3 DOF or better (e.g., focus repeatability for a 20X lens is 0.3 µm).*

b. *The Linear Range is restricted digitally by the sensor controller. The linear range setting has been found to be optimal for sensor speed and resolution. Extending linear range reduces resolution.*

c. *The specified Capture Range is valid only if WDI's coupling filter is used and no additional filters are placed between the sensor and the objective lens. Additional filters reduce light transmitted and received by the sensor and adversely affect the Capture Range.*

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Intended use

The ATF 6.5 sensor is intended to be used as a component within a user-supplied system that contains:

- an infinity-corrected microscope system;
- a Z-axis motion system used for focusing (i.e., Z-stage); and
- a computer with a serial interface.

NOTE: *WDI takes no responsibility for poor performance or malfunction of the system if the conditions described in this section are not met.*

The ATF 6.5 sensor is intended to be operated and stored under the following conditions:

1) In a non-corrosive clean room or laboratory environment.
2) In a temperature- and humidity-controlled environment:
   - Working temperature range of 18ºC to 35ºC
   - Humidity range of 10% to 75% non-condensing
   - Storage temperature range of 10ºC to 40ºC
   - Transport temperature range (in original hermetic packaging) of -20ºC to 50ºC
3) In a mechanically quiet environment (i.e., without vibration) to avoid damaging optical components.

Laser classification

Both the 680 nm and 785 nm versions of the ATF 6.5 sensor were tested and comply with EN 60825-1 and IEC 60825-1 for a Class 3B laser product.

Although the ATF sensor contains an embedded Class 3B laser component and may be rated as a Class 3B laser product, when properly integrated as part of a complete system and used as intended, the system rating is a Class 1 laser product. This is because the standards are concerned with laser radiation exposure to the user within the scope of the intended use of the product, and the intended use of this product is to be used as part of a complete system.

The ATF sensor is attached to the microscope through an optical port perpendicular to the infinity range of the microscope. A coupling filter then directs the beam through the microscope objective. For proper operation, the sensor aperture must be at, or slightly inside, the optical port and the optical port must be relatively light tight. This means there should be no laser radiation leakage at this coupling. In addition, since the specimen is on a Z-stage that is moved up or down to bring the specimen into focus, it would be impractical for the user to view the laser beam at the focal point. Thus, any laser radiation exposure to the user would be from a reflected beam through the objective lens. Testing has shown that this level of exposure is eye safe and the complete system can be rated as Class 1.

CAUTION: It is the integrator's responsibility to ensure that there is no laser light leakage at the optical port. For details, see "Optical port" on page 25.

Regulatory compliance

The ATF 6.5 sensor equipment has been tested and found to comply with the limits of a Class A digital device in accordance with Part 15 of the FCC rules and regulations found in Section 15.105. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

NOTE: *If the product is modified or changed, the FCC and/or CE approval becomes invalid. In this case, you are responsible for ensuring product conformity.*

The equipment generates and radiates radio frequency energy and, if not installed and used in accordance with this user guide, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at the user's own expense.

Information to user (FCC Section 15.21) – The user is cautioned that any changes or modifications not expressly approved by WDI or an authorized representative could void the user's authority to operate the equipment.

Special accessories (FCC Section 15.27) – The user is cautioned that any peripheral device installed with this equipment, such as a computer, must be connected with a high-quality shielded cable to ensure compliance with FCC limits.

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Changes to WDI products

WDI reserves the right to improve, change, or modify products without incurring any obligations to make changes to previous WDI equipment.

Warranty policy

WDI warrants each new customer product to be free of defects caused by faulty materials or poor workmanship for a period of 24 months from the date the system was installed by WDI or an authorized dealer or agent.

The Warranty Policy does not apply to defects caused by abnormal events or misuse, improper maintenance or installation, including excessive wear, for which WDI is not responsible.

Parts and equipment replacement

It is at the sole discretion of WDI to repair or replace any unit or part that fails within the limits of the Warranty Policy.

The Warranty Policy is valid only on new equipment. Replacement of parts, components, and/or PCB boards during a warranty repair does not extend the original warranty period.

Consumable materials, such as fuses, are not covered by the Warranty Policy.

Warranty repair process

Key components of WDI products are relatively small and are not field repairable. Warranty repairs are performed at WDI or authorized service locations.

When requesting warranty service, customers must obtain a Return of Materials Request (RMA) from WDI, and provide the following information:

- Purchase Date
- Problem Description
- Equipment Model Number and Serial Number

Labor and all other costs for the removal of the WDI component being returned for repair and subsequent replacement at the customer's site are not covered by the Warranty Policy. Removal is provided by the customer or system provider, or an appropriate dealer.

Disclaimer

Maximum liability shall not, in any case, exceed the contract price of the products claimed to be defective. Consequent damages including, but not limited to, any loss of profit, property damage, or personal injury are not covered by the Warranty Policy.

Technical support

For technical support, please contact our Applications Team at *applications@wdidevice.com*.

Safety notices

In this document, cautions and warnings are defined as follows:

- **Warning** – Identifies a danger that could result in personal injury or death.
- **Caution** – Identifies a hazard that could result in damage to equipment or loss of data.

WARNING! Do not open the sensor's cover.

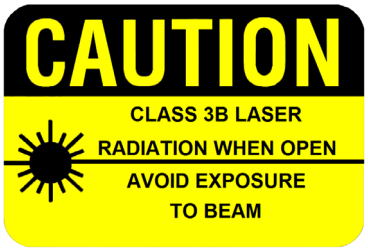
WARNING! Do not stare into the laser beam.

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Warning labels

WDI products have labels affixed to their packaging or enclosures, similar or identical to those shown below.

WARNING! Be sure to read and follow all warning labels.



The product contains a Class 3B laser component within the sensor casing. A Class 3B laser can cause eye injury. Do not open the sensor casing. If opened, you can be exposed to radiation from the laser beam.



The product contains a laser beam. Avoid direct exposure to the beam.



In both the visible and invisible wavelength range, the laser radiation from the sensor output port is Class 3B. Avoid direct exposure to the beam.



Electrostatic discharge (ESD) can damage or destroy the product's electronic components. Observe precautions for handling these components (e.g., use anti-static mats, gloves, wrist straps).

CHAPTER

2

Setting up the system

This chapter describes how to install the ATF 6.5 sensor on the microscope system and connect the components. Be sure to read this chapter completely before attempting to install the system.

> ***NOTE:*** *The microscope system diagrams in this guide are provided for illustrative purposes only and may not match your system accurately.*

The following topics are covered:

- *System overview, pg. 20*
- *System components, pg. 21*
  - *MCM+ connections, pg. 32*
- *Installing the sensor, pg. 23*
  - *Typical setup, pg. 23*
  - *General installation principles, pg. 25*
  - *Optical port, pg. 25*
  - *Coupling filter, pg. 27*
  - *Mounting bracket, pg. 27*
- *Connecting the system, pg. 31*
- *Sensor status indicators, pg. 37*

System overview

Figure 1 illustrates how the ATF 6.5 sensor integrates with a simple microscope system. The block diagram shows the components and configuration for the full autofocus system (ATF SYS), which is the main focus of this user guide. Several other variations of the sensor are available for purchase. For details, see *"Sensor variations" on page 105*.

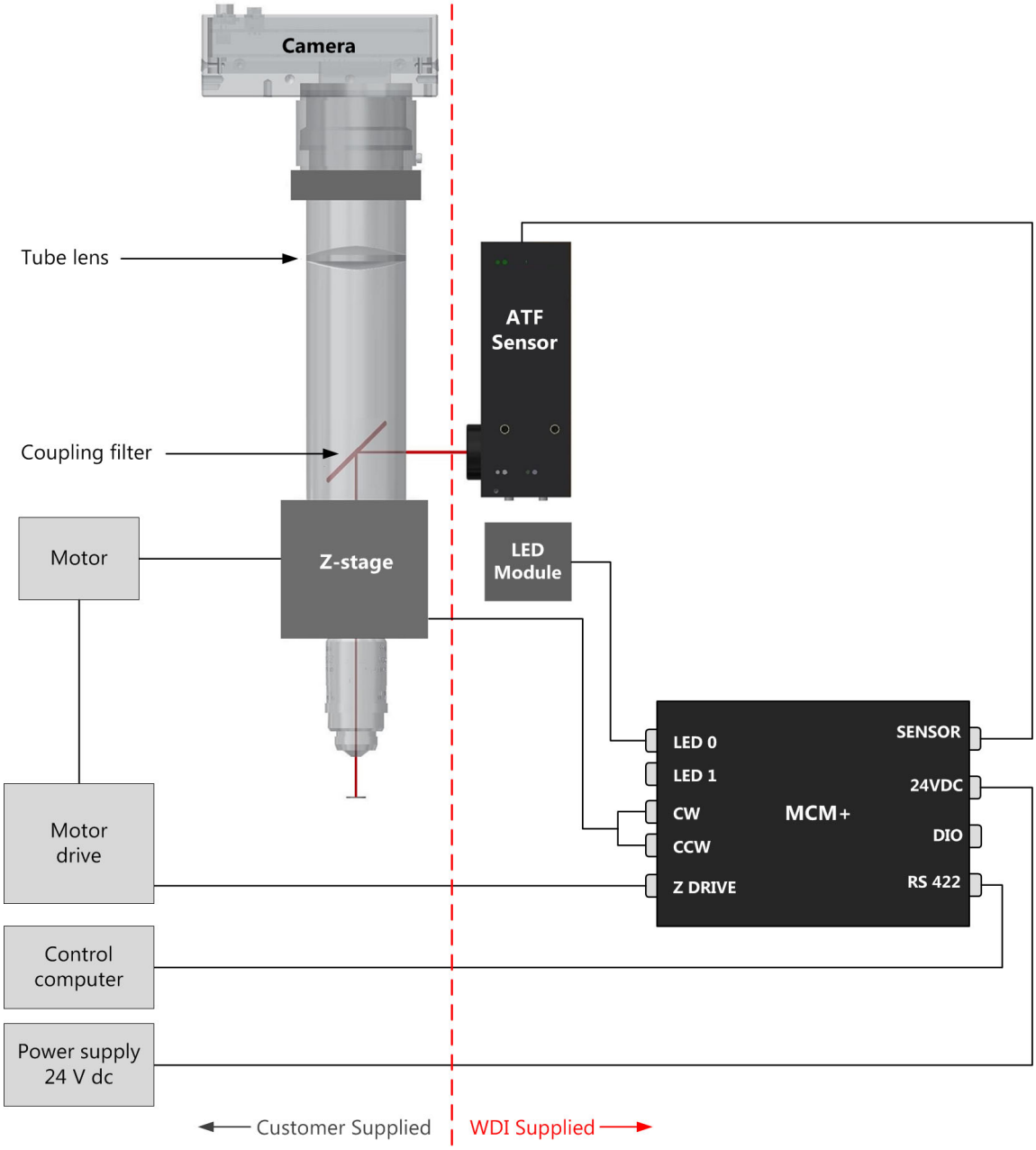


Figure 1 *System overview*

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System components

Before you begin installing the system, ensure that you have all the components and cables listed in this section.

The following system components are supplied by WDI:

- **ATF Sensor** (Part #800904) – The ATF 6.5 sensor is installed on the microscope system and connects easily with a single cable. For installation instructions, see *"Installing the sensor" on page 23*. For connection details, see *"Sensor connection" on page 33*.



Figure 2 *ATF 6.5 sensor*

- **MCM+** (Part #800902) – Each ATF sensor is shipped with an MCM+ (motor control module), which functions as an auto-focusing servo controller. For connection details, see *"MCM+ connections" on page 32*.



Figure 3 *MCM+*

- **Sensor cable** (Part #800702) – Connects the ATF 6.5 sensor to the MCM+. For connection details, see *"Sensor connection" on page 33*.



- **COM cable** (Part #800903) – Connects the MCM+ to the control computer serial port. For details, see *"COM connection" on page 33*.

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- **MCM+ Connector Kit** (Part #800662) – Provides connectors for cabling of the following:
  - 24 V dc power supply (see *"Power connection" on page 33*)
  - Digital input and outputs (see *"DIO connections" on page 33*)
  - Z-drive (see *"Z-drive connections" on page 35*)
  - Limit switches (see *"Limit switch connections" on page 36*)
  - LED for camera illumination (see *"LED illuminator connections" on page 36*)

  The connectors are provided without cables because the cable type and length depend on the application and the equipment to which they will be connected.

The following user-supplied components are also required:

- **Power supply** – A 24 V dc, 3A (minimum) power supply must be supplied for standard applications.
- **Z-stage system** – A system for motorized Z-axis focusing must be provided, complete with Z-stage, motor drive, and motor. It must accept step and direction signals from, and provide CW and CCW limit switch signals to, the MCM+.
- **Control computer** – The software used to configure and operate the ATF 6.5 sensor and Z-stage system must be installed on a user-supplied computer. Ensure that the computer has a grounded power cable.
- **LED illuminator** – At least one LED module is normally used with the system for illumination. You can purchase the LED module from WDI.

Installing the sensor

Installation of the ATF 6.5 sensor depends on the system application and equipment. Most installations can be completed easily by following the instructions provided in this section.

Typical setup

When the ATF 6.5 sensor is used in a simple microscope setup, as shown in *Figure 4*, the sensor projects the laser beam through the objective lens and onto the target. Reflection from the target passes back through the objective lens, through the coupling filter, and then back to the sensor where it is processed. An internal ATF imaging sensor captures the returned light, which enables calculation of the distance to the best focal plane.

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In general, the setup must ensure that:

- The microscope's objective lens is coaxial with the review camera and tube lens.
- The microscope's main optical axis is perpendicular to the objective plane and camera image plane.
- The review camera is coaxial with the microscope's optical axis.

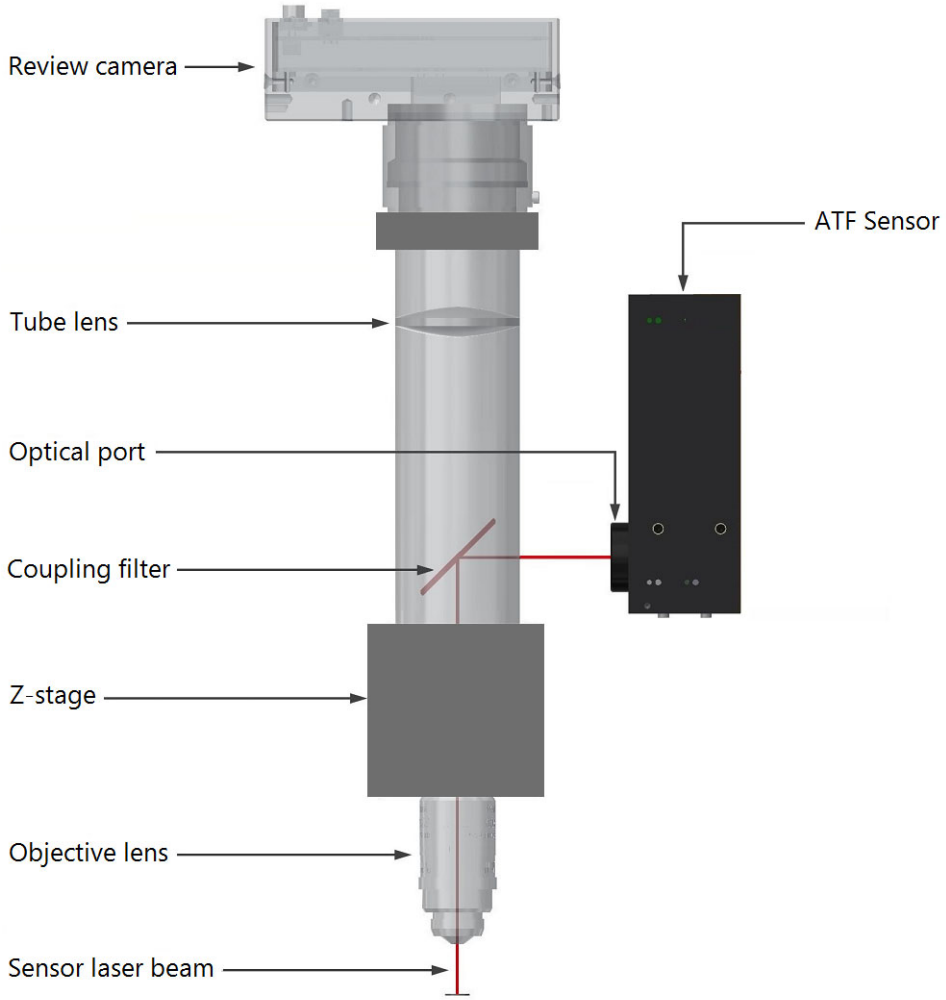


Figure 4 *Typical microscope setup*

If autofocus is enabled, the sensor provides continuous motion corrections to a stepper motor via the MCM+ resulting in actuation of the Z-stage to eliminate focus error.

The sensor operates best when the laser line width is the thinnest. To prevent aberrations in the images, the laser beam must pass through the objective lens coaxially.

When the distance between the objective lens and the sensor is less than 100 mm, angular adjustment is generally not required and reliance on machining tolerance is adequate. If the

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distance is greater than 100 mm, a coupling filter angular adjustment is required to correct the small angular error that may create a large offset of the laser beam position at the objective lens.

General installation principles

Although each installation is different, the following general principles must be adhered to regardless of the setup.

Minimize stray light and stray reflections

The sensor operates by analyzing the reflection of its own laser beam off the target sample. When stray light from other sources enters the sensor, it can affect the sensor's accuracy and range. One of the main causes of stray light is stray reflection.

Stray reflections can come from other optical components within the system that are:

- not properly coated with anti-reflection coating;
- not properly aligned; or
- have reflective surfaces (e.g., glass or aluminum placed at 90° to the optical axis).

NOTE: *If your setup requires a surface to be positioned 90° to the optical axis, ensure that the surface is non-reflective, rough, or has a small radial curve.*

Align objective lenses coaxially

If your system has a lens-changing mechanism, ensure that it is configured so that all objective lenses come to rest at a position that aligns their optical axis with the primary optical axis.

NOTE: *Misaligned lenses will cause the sensor to function sub-optimally and can potentially cease operation.*

Optical port

The ATF 6.5 sensor operates at an extended range of up to 300 mm from the microscope objective lens back focal plane. However, minimizing this distance will result in capture range gain at high NA lenses and will simplify sensor alignment.

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The sensor should be mounted on the microscope such that the sensor's laser beam enters the microscope at an optical port that is close to the objective lens, passes through the objective lens, and is reflected by the target.

The selected optical port should have a slot slightly wider and 5 mm longer in each direction than the sensor's aperture. This will ensure that the sensor's aperture is mounted directly inline with, or extended slightly into, the optical port so that the laser beam is contained within the microscope body and stray light is excluded from the optical path. For details, see *Figure 5*.

For the ATF 6.5 sensor to operate at full capacity, the optical port must meet the following criteria:

- Be as close as possible to, but no more than 300 mm from, the objective lens.
- Allow the sensor's laser beam to follow the optical axis.
- Allow space for the sensor’s electrical connections.

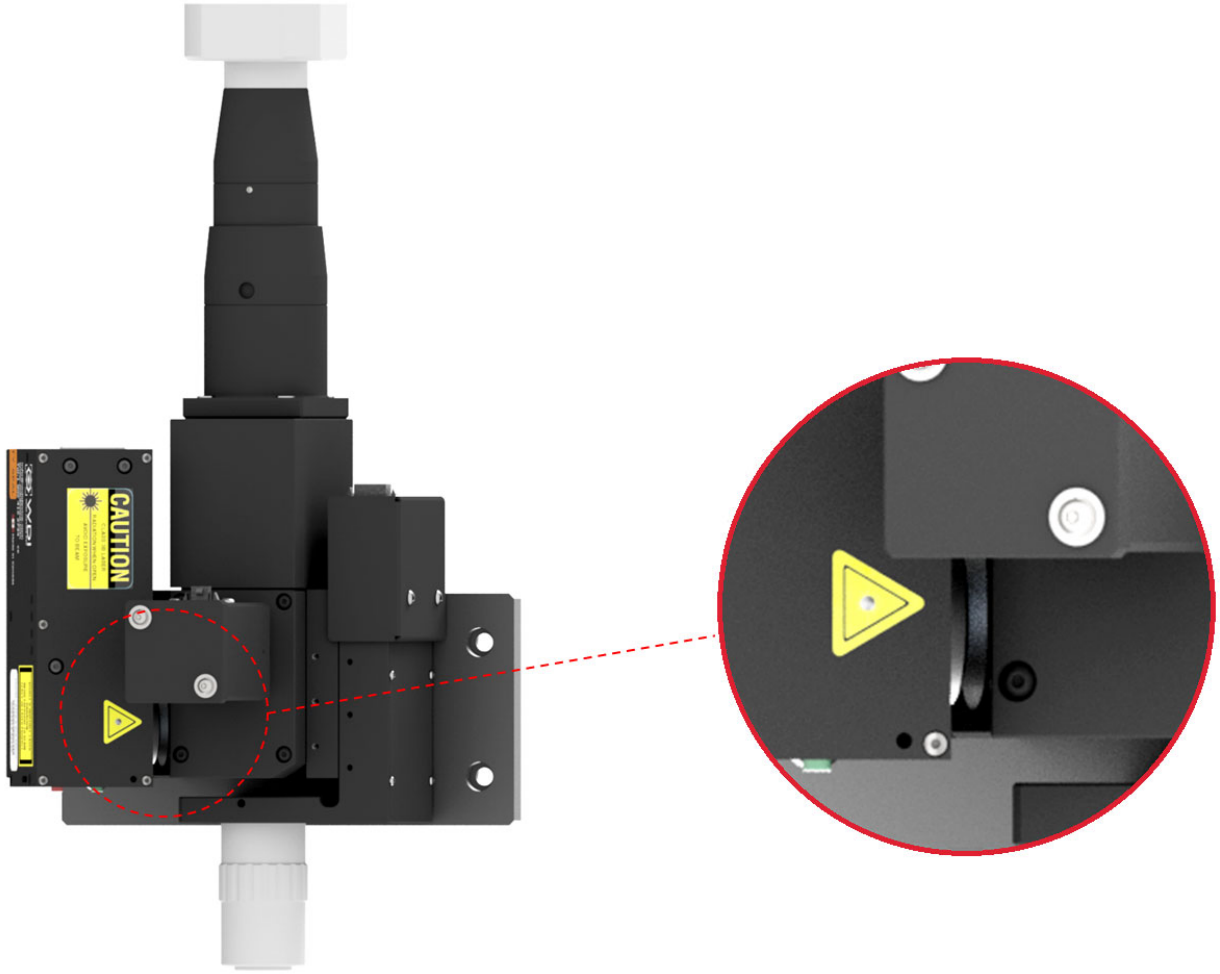


Figure 5 *Microscope optical port*

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Coupling filter

The ATF 6.5 sensor requires the use of an optical coupling filter that is installed inline with the selected optical port. This component is crucial in establishing the optical path of the sensor's laser, and also helps to reduce the amount of noise reaching the sensor. The coupling filter is mounted in the infinity range of the microscope no further than 300 mm from the objective lens back focal plane.

The most important characteristic of the coupling filter is that it reflects as much laser light as possible back into the sensor without reflecting too much ambient light. For this reason, we recommend the use of a notch filter, which selectively reflects light of 680 nm or 785 nm (depending on your sensor's laser wavelength) while transmitting all other wavelengths. If a beamsplitter is used, it should have a T/R appropriate for the intended target. For some specific video applications, a beamsplitter with a T/R of 50/50 can be used.

The coupling filter should also meet the following requirements:

- Is optimized for a 45° angle of incidence. This matches the angle at which the sensor's laser beam is incident on the coupling filter in a typical setup.
- Has a dielectric coating on the front surface to improve long-term stability.

NOTE: *While it is possible to use other filter types (e.g., bandpass filter, neutral density filter), we recommend the use of a notch filter. If you would like to use a different type of filter, please consult with our Applications Team at applications@wdidevice.com.*

Mounting bracket

A mounting bracket is required to mount the sensor to the microscope system, similar to the bracket shown in *Figure 6*. To facilitate attaching the sensor to the mounting bracket, the sensor has two sets of four threaded holes in standard M2.5 and M3 size, respectively, that are separated by a distance of 105 mm (see *Figure 7*). You can choose the appropriate size screws for your setup.

CAUTION: The mounting screws must not penetrate the sensor's housing more than 5 mm deep. Longer screws may damage the sensor or irreversibly upset its internal alignment.

Ensure that the mounting bracket is compatible with both your microscope setup and the sensor mounting holes. The mounting bracket must also allow the sensor to be adjusted 5 mm along the length of the sensor's body in either direction. This is typically done using slotted mounting holes on the bracket.

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The sensor operating temperature range is 10 to 35 °C. Cooling is provided mainly through the sensor body to the mounting bracket. If the sensor casing temperature exceeds 35 °C, forced air cooling is recommended.

For initial installation, position the sensor so that the mounting screws are in the middle of the slotted mounting holes.

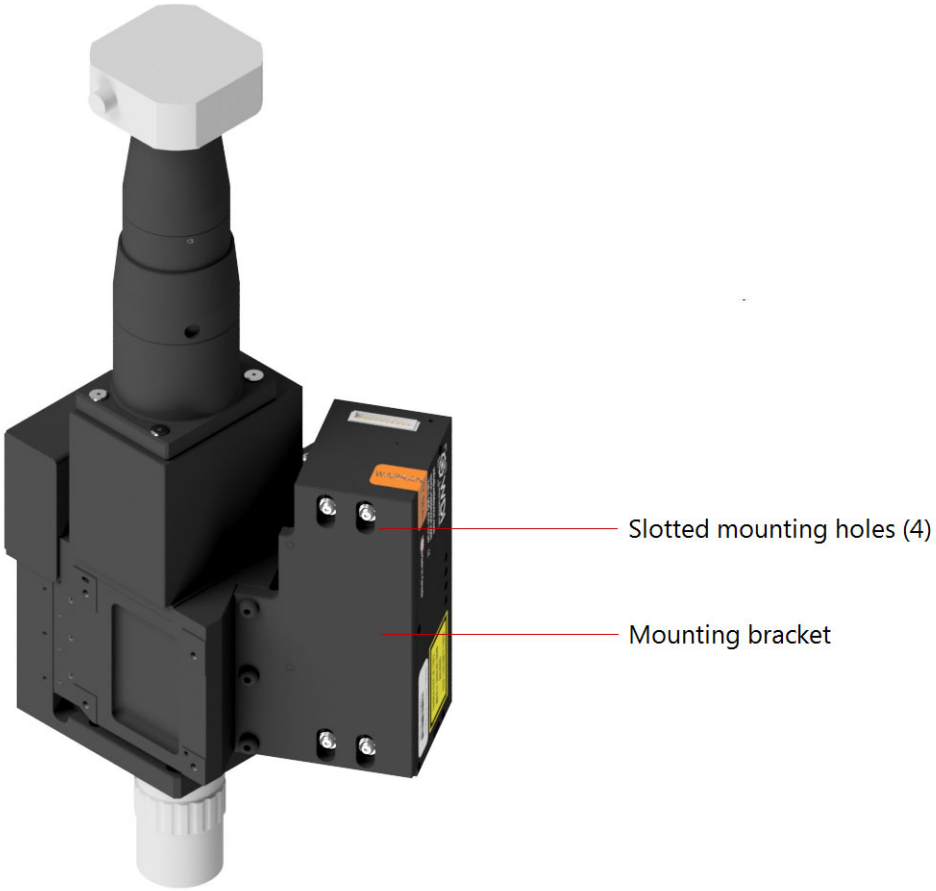


Figure 6 *Sensor mounting bracket*

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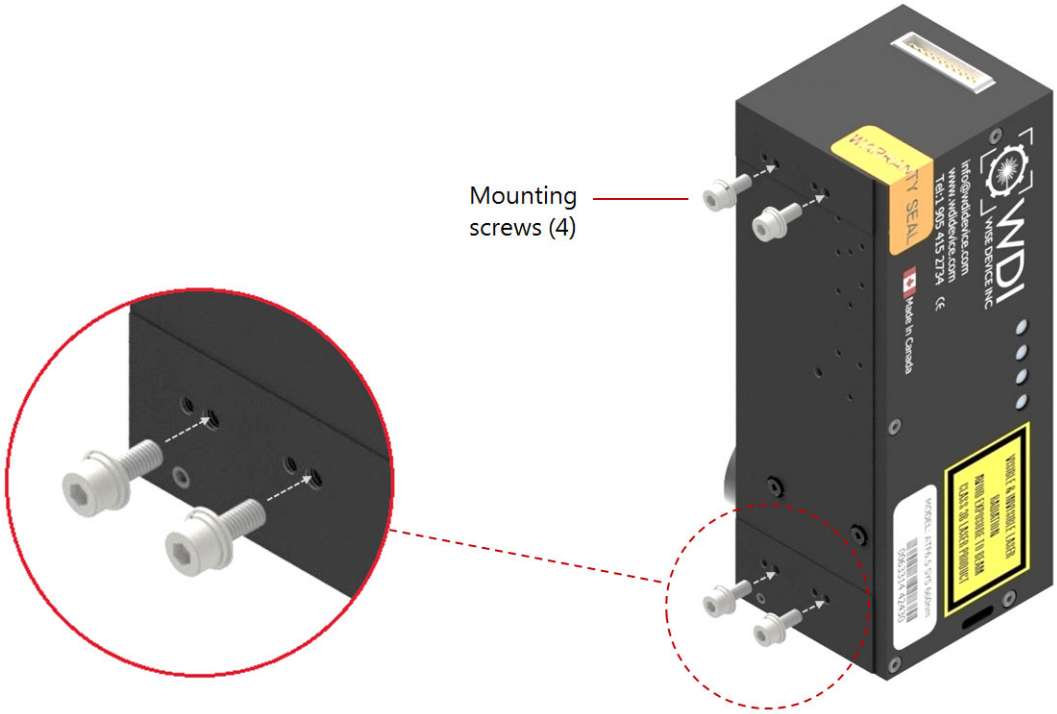


Figure 7 *Sensor mounting screws*

Space requirements

As shown in *Figure 8*, when you install the ATF 6.5 sensor, you must allow at least 55 mm of space at both ends of the sensor to accommodate the sensor cable connector at one end and to allow access to the adjustment screws at the other end.



Figure 8 *Sensor space requirements*

Review camera

When installing the review camera, ensure that the camera is aligned with respect to the microscope's optical axis.

Connecting the system

Most of the system components are connected through the MCM+, as shown in *Figure 9*. For details on the connections, see *"MCM+ connections" on page 32*.

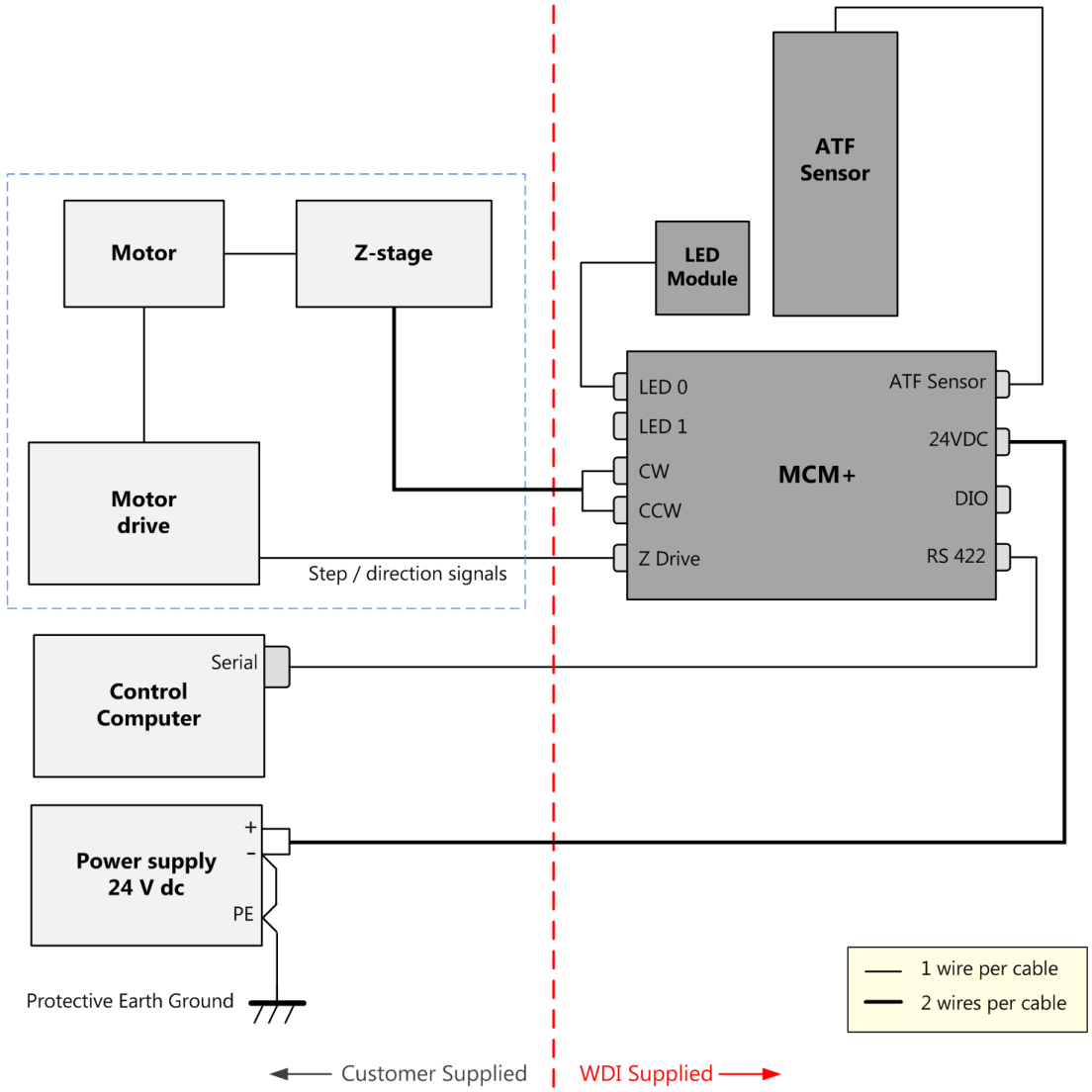


Figure 9 *Connection block diagram*

The system components must be grounded. The superstructure can pick up electromagnetic radiation from high power and high frequency sources. If not adequately grounded, the electrical noise and ground shift can degrade system performance. Connecting the components to protective earth ground (PE) is usually enough to prevent unwanted EMI and ground shifts from disturbing the system.

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MCM+ connections

The ATF 6.5 sensor is shipped with an MCM+, which is used to control the motor-driven Z-axis of the microscopy payload. This section describes the MCM+ connections. For complete details on how to use and configure the MCM+, refer to *UM800046 MCM+ User Guide*.

Some applications might require a different model controller. For example, an MCZ is required when the system is integrated with a ZAA manufactured by WDI. An MFC is required when the system is using an optional lens changer, similar to the LLC manufactured by WDI. In most cases, the connections are very similar to the MCM+. For details, refer to the appropriate controller's user guide.

The MCM+ connections are as follows:

- **SENSOR** – 12-pin connector for the sensor cable.
- **24VDC** – 24 V dc power for the controller.
- **DIO** – Digital input/output for the E-Stop, C-Sync, In-Range and In-Focus signals.
- **RS 422** – COM connection to the control computer serial port.
- **Z DRIVE** – Step and direction outputs for the Z-stage system.
- **CW** and **CCW** – Limit switch inputs.
- **LED 1** and **LED 0** – LED output for camera illumination.

For details on these connections, refer to the following sections.

NOTE: *For the DIO, LED 1, and LED 0 connectors, Pin 1 is on the left. For all other connectors, Pin 1 is on the right.*



Figure 10 *MCM+ connections*

Sensor connection

The ATF 6.5 sensor input and output are transmitted through a single HRS-3500 port. The sensor comes with a sensor cable (Part #800702) that terminates with a Hirose HRS-3500 connector at the sensor end and a 12-pin connector at the controller end.

Power connection

The power input supplies the MCM+ with 24 V dc power, and consists of two pins that are connected as shown in *Table 3*. You must assemble the cable using the 24 V dc connector provided in the MCM+ Connector kit (Part #800662). We recommend using a 22 AWG shielded cable.

Table 3 24VDC power connector

Connector	Pin	Signal	Function
24VDC	1	+24 V dc	Power for MCM+
	2	0 V	Connected to protective earth at power supply end

COM connection

The MCM+ has an RS-422 port for communication with the control computer. The RS-422 cable (Part #800903) is a 1.8 meter, straight-through serial extension cable with an RJ11 connector at both ends. The cable connects to the RS-422 port on the controller, and to the serial port on the control computer via an RS-232 to RS-422 converter (Part #800276). For convenience, we recommend that you label the serial port on the control computer that is used for this purpose.

Some applications might use optional RS-232 communications, in which case an RS-232 cable (Part #800004) is provided. The 1.8 meter RS-232 cable has an RJ11 connector at the controller end and a DB9 connector on the computer end.

DIO connections

This section describes the MCM+ digital input/output connections. The pin assignments are provided in *Table 4*.

NOTE: *The E-Stop input (pins 1 & 2) on the DIO connector must be supplied with power for the ATF 6.5 sensor and Z-stage to function.*

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There are two digital inputs and two digital outputs as follows:

- **E-Stop input** – Used for the emergency stop safety feature. While a signal is applied, the ATF 6.5 sensor and Z-stage are active. If the signal is interrupted, all functions, including laser power and Z-stage motion, are immediately stopped.
- **C-Sync input** – Used for the optional camera sync feature, which temporarily disables the sensor's laser while an image is being captured to prevent the laser line from appearing on the image. The C-Sync input can be driven from the output of an opto-coupler, as shown below.

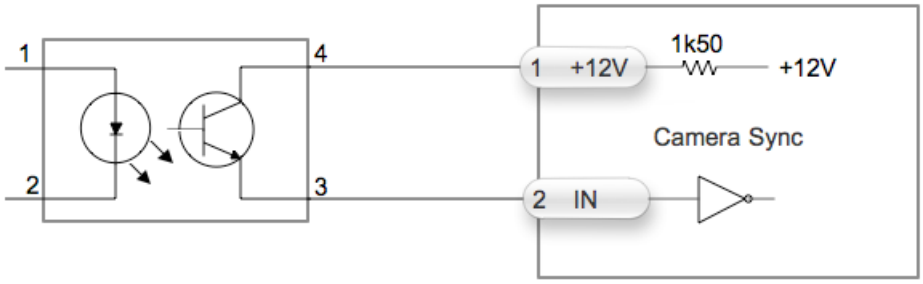


Figure 11 *C-Sync wiring*

- **In-Range output** – This optional output is active when the sensor detects a sample within its capture range. It is an open collector type output that is active when its voltage is 1 V or less, and can be used to drive an opto-coupler, as shown below.
- **In-Focus output** – This optional output is active when the sensor detects that the sample is at the best focus position. It is the same type as the In-Range output described above.

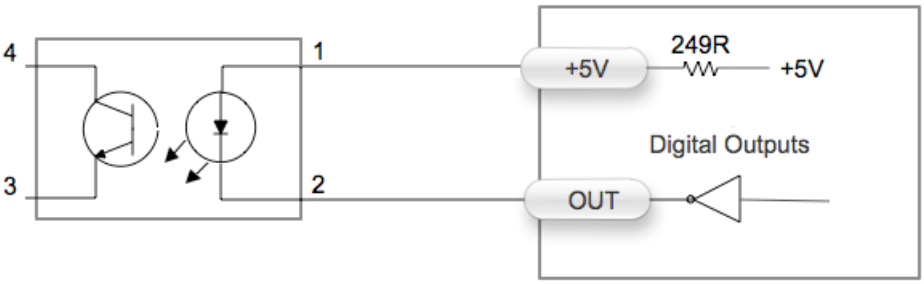


Figure 12 *In-Range and In-Focus wiring*

Table 4 DIO connector pin assignments

Connector	Pin	Signal	Function
DIO	1	+12 V	Power for E-Stop input
	2	E-Stop	Digital input
	3	+12 V	Power for C-Sync input (optional)
	4	C-Sync	Digital input (optional)
	5	+5 V	Power for In-Range output (optional)
	6	In-Range	Digital output (optional)
	7	+5 V	Power for In-Focus output (optional)
	8	In-Focus	Digital output (optional)

Z-drive connections

The motor drive and motor control the Z-stage. For pin assignments, see *Table 5*.

Table 5 Z DRIVE connector pin assignments

Connector	Pin	Signal	Function
Z DRIVE	1	0 V	Z-drive phase cable shield (do not terminate the shield at the motor end)
	2	A+ Phase	Z-drive phase wiring
	3	A- Phase	
	4	B+ Phase	
	5	B- Phase	

Limit switch connections

The CW and CCW limit switch connectors provide pins for +5 V, limit switch output, and 0 V. These inputs are intended for a high side limit switch that sources current when not triggered. For pin assignments, see *Table 6*.

Table 6 Limit switch connector pin assignments

Connector	Pin	Signal	Function
CW	1	+5 V	Power for CW limit switch
	2	CW Limit	CW limit switch output
	3	0 V	CW limit switch ground
CCW	1	+5 V	Power for CCW limit switch
	2	CCW Limit	CCW limit switch output
	3	0 V	CCW limit switch ground

LED illuminator connections

The LEDs provide camera illumination. For most applications, only one LED is used; however, a second LED can be used, for example, to provide illumination from the bottom or in a different color. You can purchase an LED from WDI, or you can provide your own. For pin assignments, see *Table 7*.

Table 7 LED illuminator pin assignments

Connector	Pin	Signal	Function
LED 0	1	V+	LED positive voltage (anode)
	2	V-	LED negative voltage (cathode)
LED 1	1	V+	LED positive voltage (anode)
	2	V-	LED negative voltage (cathode)

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Sensor status indicators

ATF 6.5 sensor has four status indicators, as shown in *Figure 13*. When the sensor is powered up, the indicators light up and are then deactivated by the software to prevent any extraneous light from affecting the setup. During normal operation, the functions are as described in *Table 8*.

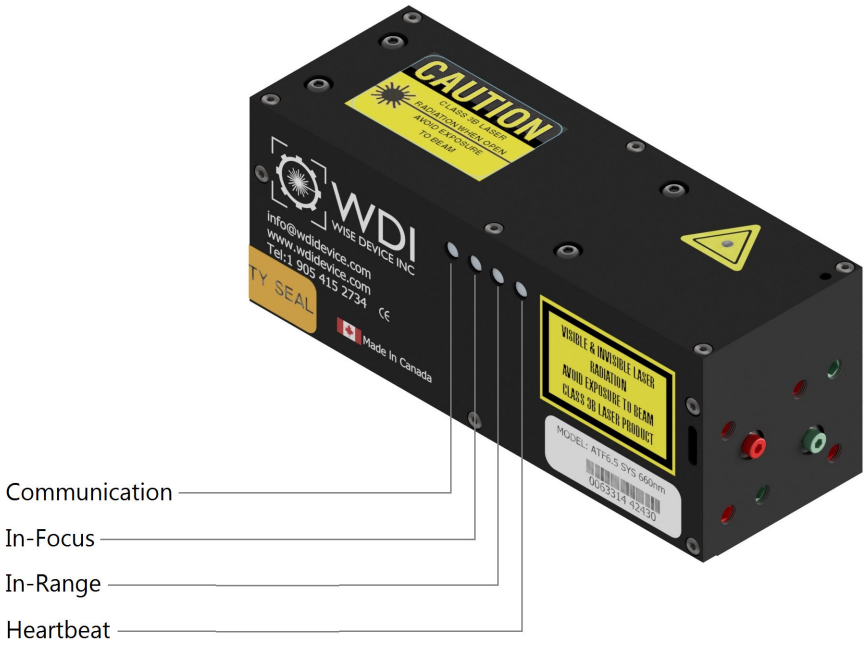


Figure 13 *Sensor status indicators*

Table 8 Sensor status indicator functions

Indicator	Function
Communication	• Blinks when serial communication is exchanged. • Blinks when memory is written or erased. • Blinks during boot time.
In-Focus	• Blinks during boot time. • Lights when the microscope is in focus.
In-Range	• Lights when the sensor can detect a sample.
Heartbeat	• Blinks at 0.5 second intervals when the firmware is loaded and operational. • Blinks in a sequence of two to nine pulses when the startup check routine fails. In this case, you must send the sensor back to WDI for repair. Follow the instructions provided in *"Warranty repair process" on page 15*.

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CHAPTER

3

Setting up the software

This chapter describes how to install and set up the ATF Console software, provides an overview of the main panel, and describes some of the more commonly-used software procedures.

The following topics are covered:

- *ATF Console software overview, pg. 40*
  - *Computer system requirements, pg. 40*
- *Installing the software, pg. 41*
  - *Uninstalling the software, pg. 41*
- *Main panel overview, pg. 42*
  - *Software sensor status indicators, pg. 44*
  - *Menu reference, pg. 45*
  - *Color definitions, pg. 45*
- *Setting up the ATF Console software, pg. 46*
  - *Starting the software, pg. 46*
  - *Setting up the objective lenses, pg. 48*
  - *Setting up an LED, pg. 49*
- *Adjusting the Z-stage position, pg. 51*
- *Performing manual Make 0, pg. 53*
- *Exiting the software, pg. 54*

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ATF Console software overview

The ATF Console software application is used to configure and adjust the ATF 6.5 sensor. You install the software on the system's control computer.

NOTE: *This guide provides only the software instructions required to perform the procedures provided in this guide. For a description of all the software functions available, refer to UM800233 ATF Console Software User Guide. To obtain a copy of this guide, contact the Applications Team at applications@wdidevice.com.*

Computer system requirements

The control computer on which the ATF Console software is installed must meet the following minimum system requirements:

- Windows 7 operating system or later
- 8 GB of RAM
- 50 MB of free hard disk space
- Display with 1024 x 768 pixel resolution
- RS-232 compatible interface

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Installing the software

The ATF Console software must be installed on the control computer before the sensor can be configured or used. The software is distributed in .zip format; therefore, the computer must have a program for unzipping compressed files.

The ATF Console software installer features a standard Microsoft Windows setup interface, which is simple and easy to use. You can install multiple versions of the ATF Console software on the same computer, provided that you select different installation folders.

The default installation folder is C:\Program Files(x86)\WdiDevice\ATF\ATF_Console_#.#.##, where #.#.## is the current software version number.

NOTE: *You have the option to create your own software application that can be used to configure and operate the ATF 6.5 sensor. In this case, refer to the ATF Functions Library Manual and the ATF_LIB zip file.*

To install the software:

1. Extract the contents of the **ATF Console Setup.zip** file to any folder.
2. In the extracted folder, double-click **ATF Console Setup.exe**.
3. Follow the instructions in the setup file to install the software. For step-by-step instructions, refer to *UM800233 ATF Console Software User Guide*.

Uninstalling the software

You can uninstall the ATF Console Software in one of two ways:

- Using the ATF Console Uninstall application (unins000.exe) that is located in the ATF Console installation folder.
- Using the **Uninstall a program feature** in the Windows Control Panel.

Main panel overview

When you start the ATF Console software, the main panel is displayed, as shown in *Figure 14*. The controls available on the main panel are described in *Table 9*.

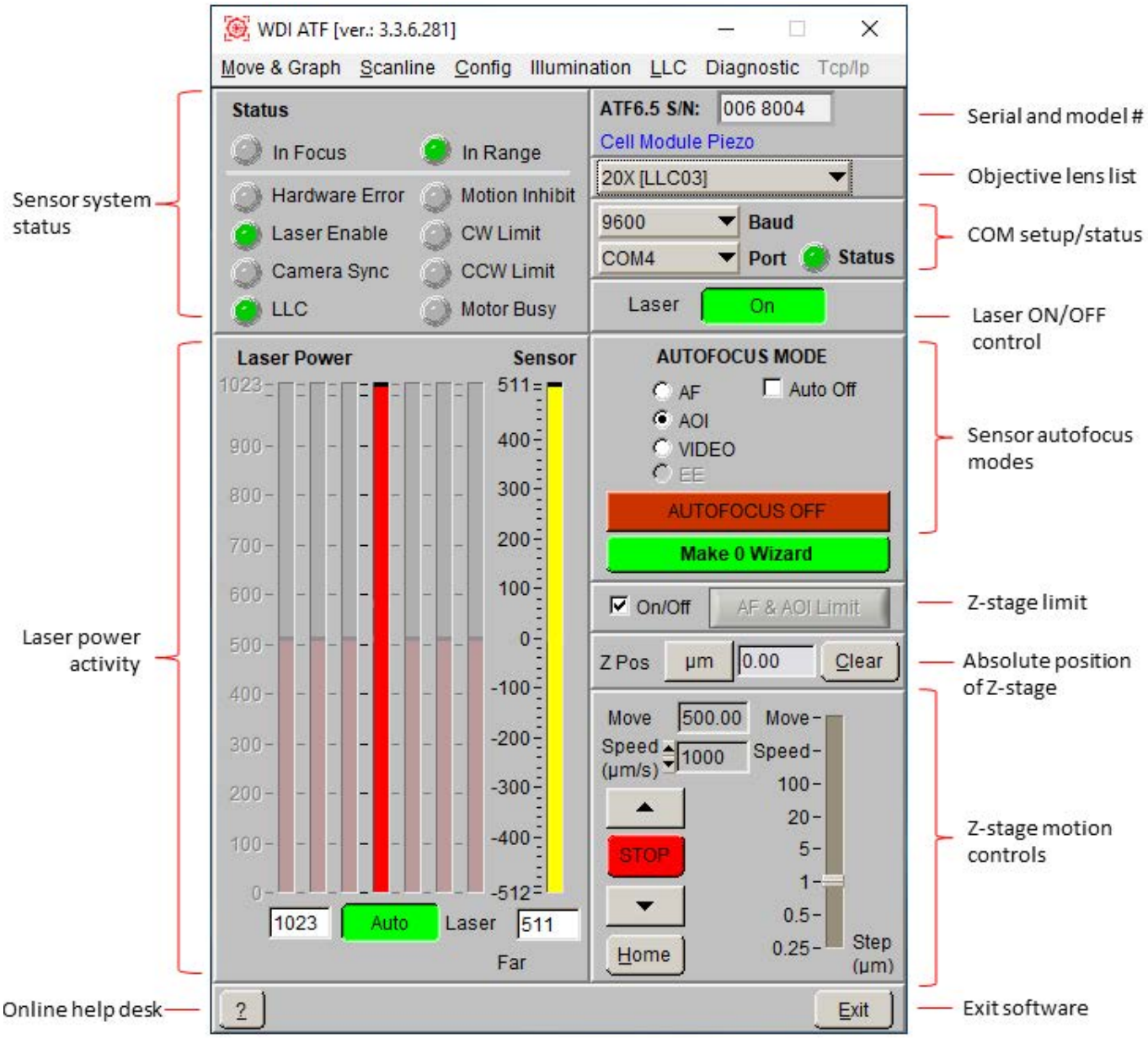


Figure 14 *ATF Console main panel functions*

Table 9 ATF Console main panel controls

Control	Description
Status	Displays the status of the sensor. For details, see *"Software sensor status indicators" on page 44*.
Serial number	Shows the sensor model and serial number. The blue text describes the sensor model (e.g., Cell Module).
Objective lens list	Allows you to select the objective lens from the list of lenses available in the system. For details on how to set up the list, see *"Setting up the objective lenses" on page 48*.
COM setup	Allows you to select the baud (i.e., transmission rate) and serial port for communication between the control computer and the sensor. For details, see *"Starting the software" on page 46*. The Status indicator turns green when communication has been established. This indicator corresponds to the Communication LED indicator on the sensor (see *"Sensor status indicators" on page 37*).
Laser	Allows you to turn the laser **ON** or **OFF**. When the laser is **OFF**, the ATF sensor autofocus is disabled. This control also acts as an indicator if the laser toggle switch is installed in the MCM+ DIO panel.
Laser Power	When in **Auto** mode, displays the laser power output set by the system. When in **Manual** mode, allows you to view and change the laser power output.
Sensor	Displays the sensor output position in Digital Units (DU), which are arbitrary values ranging from -512 to +511, as well as the sensor window mode (e.g., Near-7).
Sensor autofocus modes	Allows you to select the autofocus mode for the sensor. The laser must be turned on for the autofocus modes to be available.
AbsPos	Displays the absolute position of the Z-stage. This value is the sum of all vertical movements, both up and down (i.e., positive and negative values) of the Z-stage from the time the sensor was turned on. You can click the button to change between **µm** and **µsteps**. To reset the value to zero, click **Clear**.
Z-stage controls	Controls for moving the Z-stage (or ZAA). For details, see *"Adjusting the Z-stage position" on page 51*.
Make 0 Wizard	Accesses a wizard that is used to set the sensor autofocus zero (or reference) position.

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Software sensor status indicators

The Status indicators on the main panel display the status of the ATF 6.5 sensor. The meanings of the various status indicators are provided in *Table 10*.

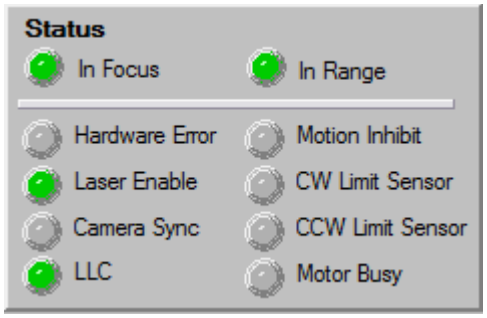


Table 10 Sensor status indicators

Menu	Description
In Focus	When green, the sensor output is within the range set in the Parameter window.
In Range	When green, the sensor is operating within the capture range and there is adequate reflected light reaching the sensor to detect the target.
Hardware Error	When red, there is a hardware problem.
Motion Inhibit	When yellow, there is an external digital input signal preventing Z-stage motion and the laser is off. Ensure that jumper installed for DIO pins 1&2.
Laser Enable	When green, the laser is turned ON. This indicator corresponds to Laser ON/OFF toggle switch.
CW Limit Sensor	When yellow, Z-stage movement is stopped by the CW limit switch. Z-stage movement is possible only in the reverse direction.
CCW Limit Sensor	When yellow, Z-stage movement is stopped by the CCW limit switch. Z-stage movement is possible only in the reverse direction.
Camera Sync	When green, sync digital input is asserted, in this example, the laser is turned off, and the autofocus activity is disabled.
LLC	When gray, no LLC is detected. When green, the LLC is detected and active from a communication point of view, but not necessarily operational (e.g., the homing operation might not have been performed). When red, the LLC is detected but there is an error.
Motor Busy	When green, the Z-stage motor is in motion.

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Menu reference

The ATF Console software menus are described in *Table 11*. For a complete description of the menus and their respective options, refer to *UM800233 ATF Console Software User Guide*.

Table 11 ATF console software menu reference

Menu	Description
Move & Graph	Validates the sensor output by plotting sensor output and laser power versus Z-position.
Scanline	Represents data from the imaging sensor. Its main functions are for sensor configuration and troubleshooting.
Config	Provides access to the sensor parameters and system configuration options. This menu is used extensively. The Parameter dialog box is accessed from this menu.
Illumination	Allows you to specify the current that is provided to each connected LED and the intensity of the illumination that it provides.
LLC	Initializes, controls, and monitors the optional Linear Lens Changer (LLC) activity, if included in the system.
Diagnostic	Collects the sensor output characteristics, hardware and software status, and sensor operating parameters. This option is available for reading files without an active sensor connected to the software console.

Color definitions

The parameters in the Parameter dialog box and a few other areas in the ATF Console software are color-coded. The colored backgrounds have the following meanings:

- **Pink** – Parameters that are associated with the selected objective lens have a pink background. Selecting a different objective lens will affect all the related parameters.
- **Light green** – Parameters that are associated with the sensor, regardless of the selected objective lens, have a light green background.
- **Blue** – Parameters that are associated with the selected substrate have a blue background. Selecting a different substrate will affect all the related parameters.

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Setting up the ATF Console software

The first time you use the ATF Console software after installing the system, you must follow the procedures provided in this section to set up the software for use with the ATF 6.5 sensor.

After the setting up the software, you can proceed to align the system for first-time use, as described in *"Aligning the system" on page 55*.

Starting the software

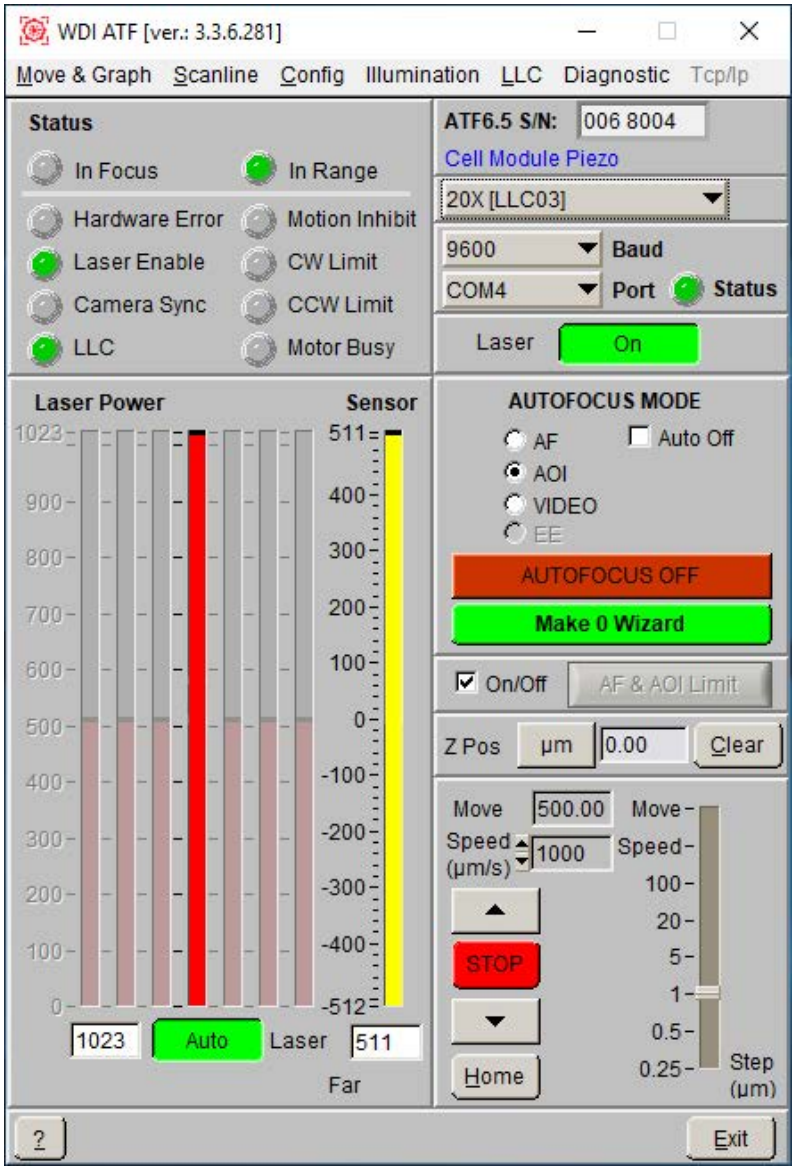
This section describes how to start the ATF Console software and establish a connection with the sensor.

NOTE: *Be sure to select the COM port before changing the baud, as described in this procedure. The baud must be set to the default value of 9600 for the initial connection with the sensor.*

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To start the ATF Console software:

1. On the **Start** menu, click **ATF_Console**.
2. On the main panel, in the **Port** list, select the serial COM port where the sensor is connected.



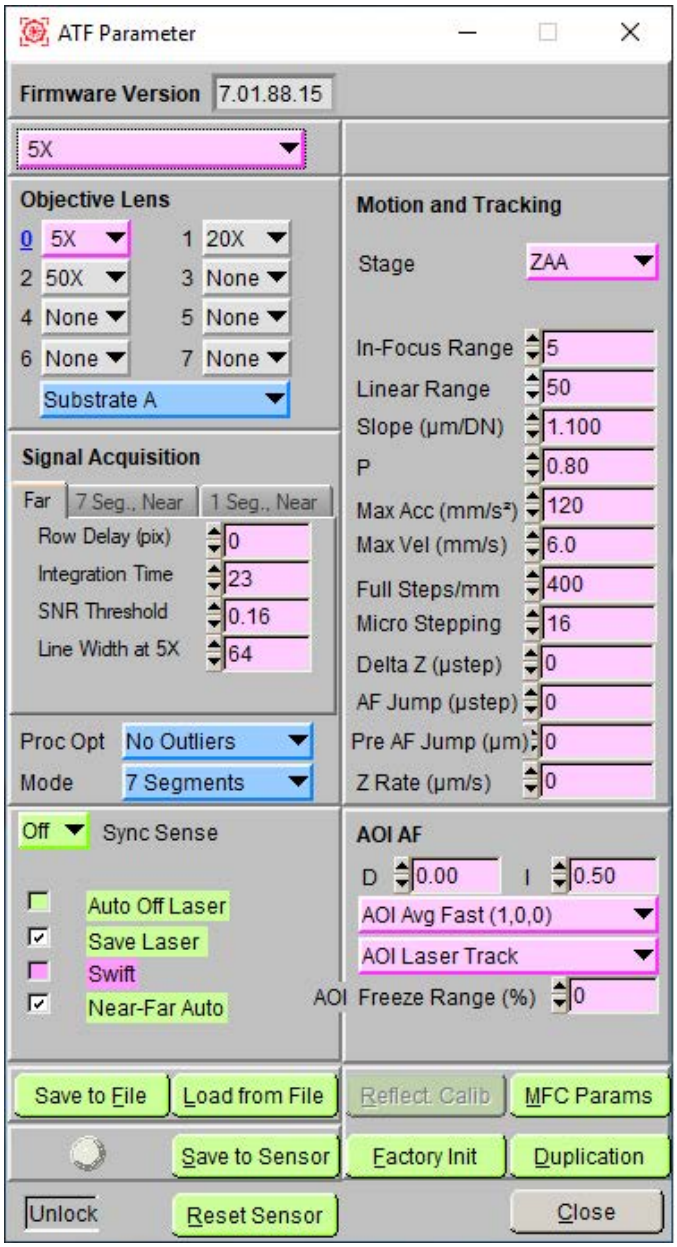
3. In the **Baud** list, select the required baud (i.e., transmission rate). Use the highest and most reliable baud possible for your installation. The baud can be affected by many factors, including COM cable length. The default baud is 9600.
4. When communication is established, the **Status** light turns green, and several dimmed controls in the main panel become available.

Setting up the objective lenses

Use this procedure to set the position of each objective lens used in the system.

To set up the objective lenses:

1. On the **Config** menu, select **Parameter**.
2. At the bottom of the **ATF Parameter** dialog box, click the **Unlock** box, type **atf6**, and then press ENTER. This will unlock the parameters, including the other options on the Config menu.



3. Under **Objective Lens**, for each objective lens used in the system, click the button next to the required position, and then select the lens from the list. For example, to set the 5X lens to position 0, click the button next to **0** and select **5X** in the list. Leave any unused positions set to **None**.
4. In the **Substrate** list, select the appropriate substrate for the target sample.
5. Click **Close**.

Setting up an LED

This procedure is used to set an LED to nominal values for the initial configuration. After the initial configuration, you can adjust the LED PWM% and current at any time for your particular objective lens and application.

You should perform this procedure for each LED used in the system. For most applications, only one LED is used. A second LED can be used, for example, to provide illumination from below (rather than above) or in a different color. If required, two additional LEDs can be connected through a second COM connection. You can purchase an LED from WDI, or you can provide your own.

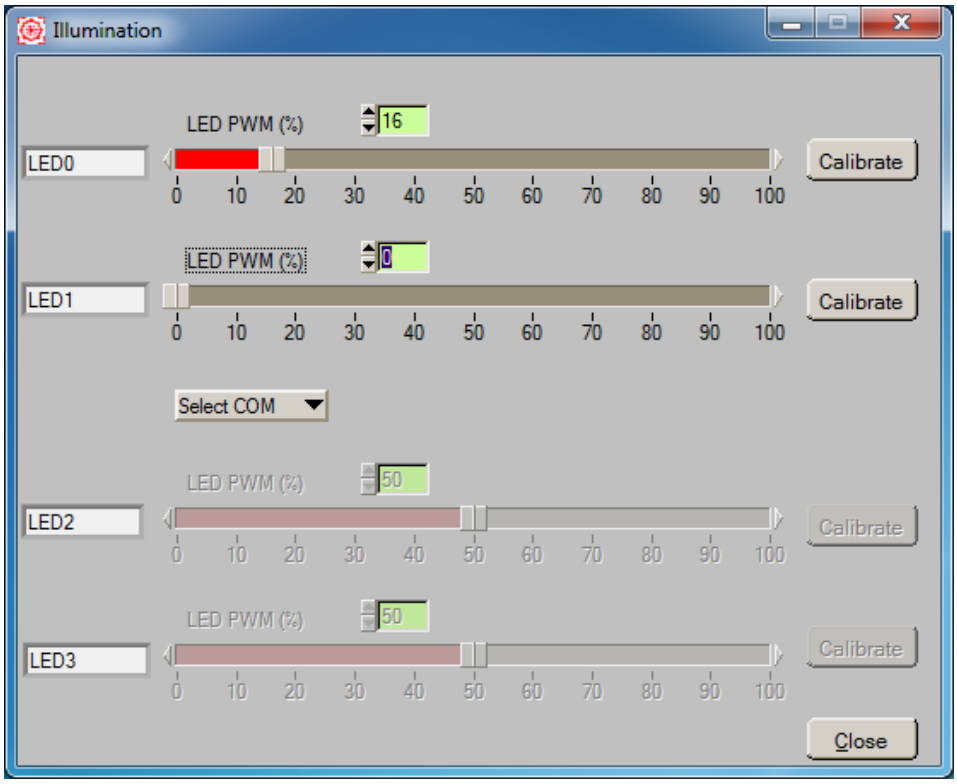
NOTE: *The procedures in this guide assume that the system is using LEDs that are supplied by WDI and controlled by the ATF Console software. If your system is using an OEM LED, you will adjust the illumination of the LED using the software provided by the LED manufacturer.*

To set up an LED:

1. Place a sample in the field of view and bring it into focus.
2. On the **Illumination** menu, click **LED**.

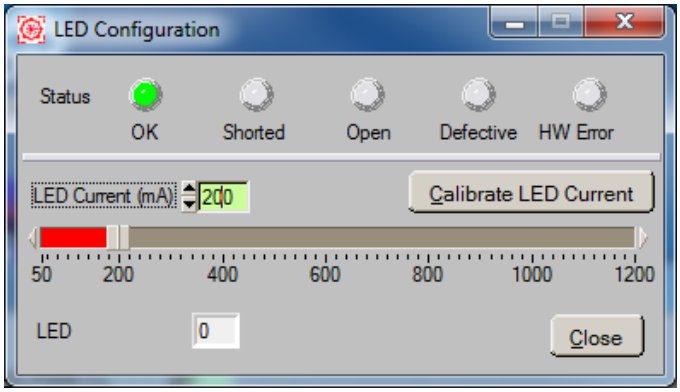
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3. In the **Illumination** dialog box, next to the LED you are configuring (e.g., **LED 0**), set the **LED PWM (%)** value to a nominal value that is appropriate for the selected objective lens (e.g., **4** for a 5X objective, **20** for 50X objective).



4. To the right of the same LED, click **Calibrate**.

5. In the **LED Configuration** dialog box, in the **LED Current (mA)** box, type **200**. This is the nominal LED current value.



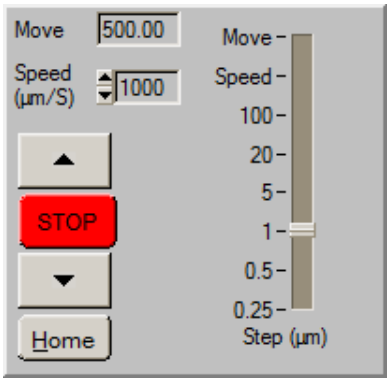
6. Click **Calibrate LED Current**. The LED should flash twice and then stay on.

7. Click **Close** to close the LED Configuration dialog box.

8. Click **Close** to close the Illumination dialog box.

Adjusting the Z-stage position

The following controls on the ATF Console main panel are used to change the position of the Z-stage. Before using these controls, ensure that the Z-stage motion parameters in the Parameter dialog box (i.e., Max Vel, Max Acc, Full Steps, and Micro Stepping) have been set correctly for your Z-stage system. For details on these parameters, refer to *UM800233 ATF Console Software User Guide*.



You can move the Z-stage in the following ways:

- In small increments from 0.25 to 100 µm.
- A specified distance.
- Continuously at a specified speed.
- To its home position.

NOTE: *To use the Home button on this panel, you must first set the home position of the sensor using the Home option on the Config menu. For details, refer to UM800233 ATF Console Software User Guide.*

To move the Z-stage in small increments:

1. Set the slider to the selected increment value from 0.25 to 100 µm.
2. Click the Up button ▲ or the Down button ▼ to move the Z-stage the specified increment in the selected direction.

To move the Z-stage a specified distance:

1. Set the slider to **Move**.
2. In the **Move** box, type the required distance in µm.
3. Click the Up button  or the Down button  to move the Z-stage the specified distance in the selected direction.

To move the Z-stage continuously:

1. Set the slider to **Speed**.
2. In the **Speed** box, type the speed at which the Z-stage will move in µm/second.
3. Click the Up button  or the Down button  to move the Z-stage continuously at the specified speed in the selected direction.
4. To stop the Z-stage while it is moving, click **STOP**.

To move the Z-stage to its home position:

1. Click **Home**.

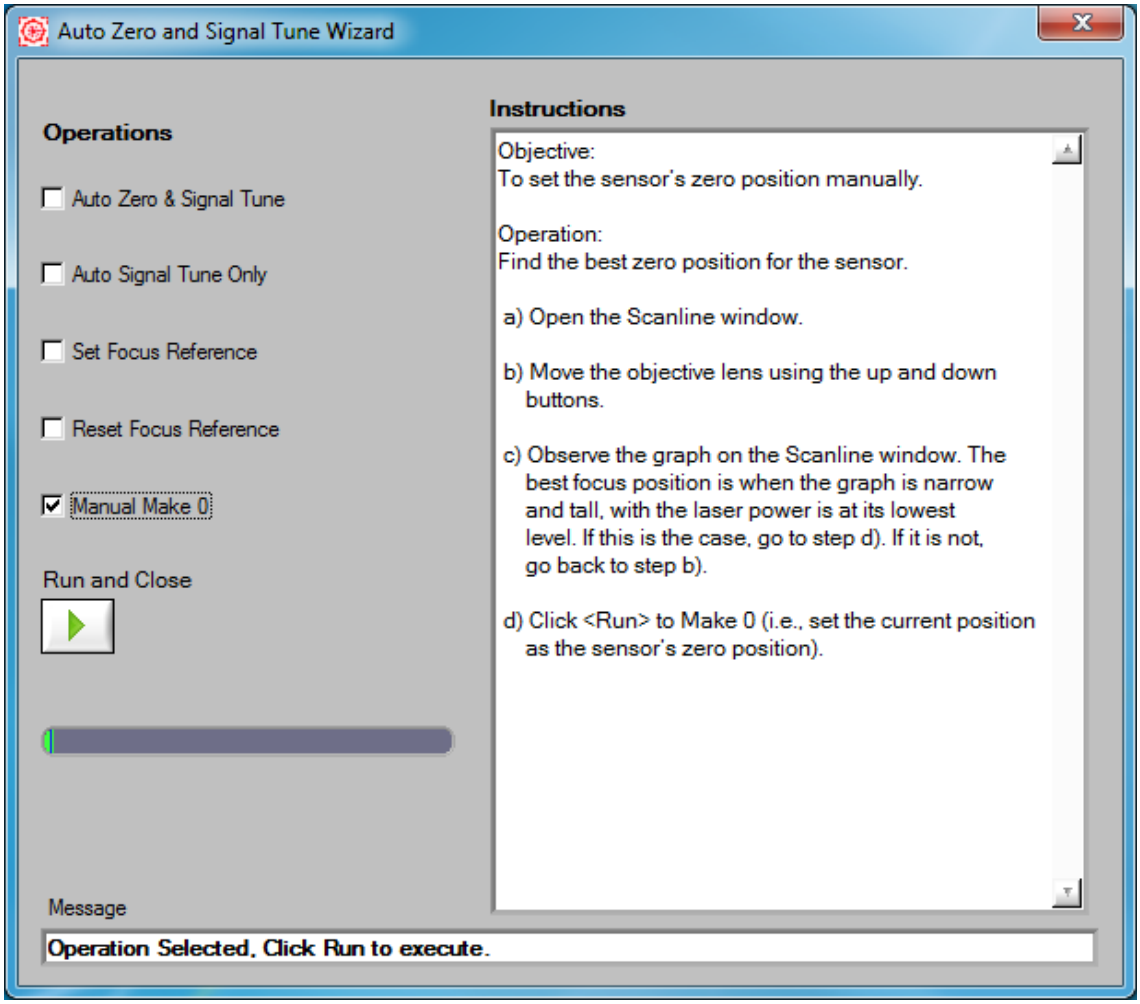
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Performing manual Make 0

This section describes how to set the current focus position as the sensor's autofocus zero (or reference) position.

To perform manual Make 0:

1. Adjust the Z-stage position to obtain a sharp, in-focus image, as seen on the camera image. For details on how to move the Z-stage, see *"Adjusting the Z-stage position" on page 51*.
2. On the main panel, click **Make 0 Wizard**.
3. In the **Auto Zero and Signal Tune Wizard** dialog box, select the **Manual Make 0** check box.



4. Click **Run and Close**. When the Make 0 operation is complete, the wizard will close.

Exiting the software

When you are finished using the system, exit the ATF Console software. It is good practice to save the sensor parameters before exiting the software.

To exit the software:

1. To save any parameters that were changed during the session, on the **Config** menu, click **Parameter**, and then do one of the following:
   - To save the current parameters permanently to flash memory on the sensor, click **Save to Sensor**. The settings will become the sensor default settings that are restored when you click **Reset Sensor**.
   - To save the current parameters for future use without making them permanent, click **Save to File**. You will be able to upload the file to the sensor at a future time.
2. Click **Close** to close the Parameter dialog box.
3. At the bottom right corner of the console, click **Exit**. The software closes.

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CHAPTER

4

Aligning the system

This chapter describes how to align the microscope system the first time after the sensor has been installed. All the procedures in this chapter must be performed in sequence for the system to work effectively and optimally for your application needs.

The following topics are covered:

- *Alignment overview, pg. 56*
- *Setting the initial sensor position, pg. 57*
- *Verifying the objective parfocality, pg. 60*
- *Centering the laser beam on the target, pg. 61*
- *Centering the laser beam on the sensor, pg. 64*
- *Aligning the half aperture for low magnification, pg. 66*
- *Aligning the half aperture for high magnification, pg. 70*

Alignment overview

You normally need to align the microscope system only once after it has been installed. There is no need to repeat system alignment unless the microscope optical components have been affected. We recommend that alignment procedures be performed in sequence by someone who has good knowledge of the microscope system and familiarity with the ATF 6.5 sensor.

Before you begin, ensure that the following system prerequisites are met:

- The microscope system is set up as described in *"Typical setup" on page 23*.
- The software is set up as described in *"Setting up the ATF Console software" on page 46*.
- The review camera is roughly positioned at a distance matching the tube lens focal length.
- The camera is activated and the camera viewer application is presenting live images.
- There is no narrow bandpass filter in front of the review camera. If a filter is present, remove it temporarily.
- The Z-stage motion parameters in the ATF Console software (i.e., Max Vel, Max Acc, Full Steps, and Micro Stepping) are set correctly for your Z-stage system.

For this procedure, you will need a specular target sample, such as polished silicon, metalized glass, or clear glass. A silicon target sample with the following characteristics is preferred:

- Free from dirt and blemishes
- Uniform height
- Highly reflective
- Contains some visible features that will help identify the best focus plane, but do not cross or disturb the laser beam line

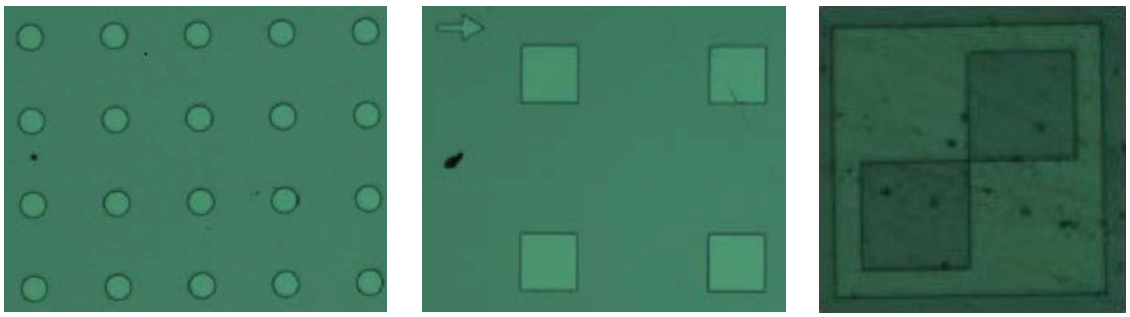


Figure 15 *Acceptable Si wafer samples*

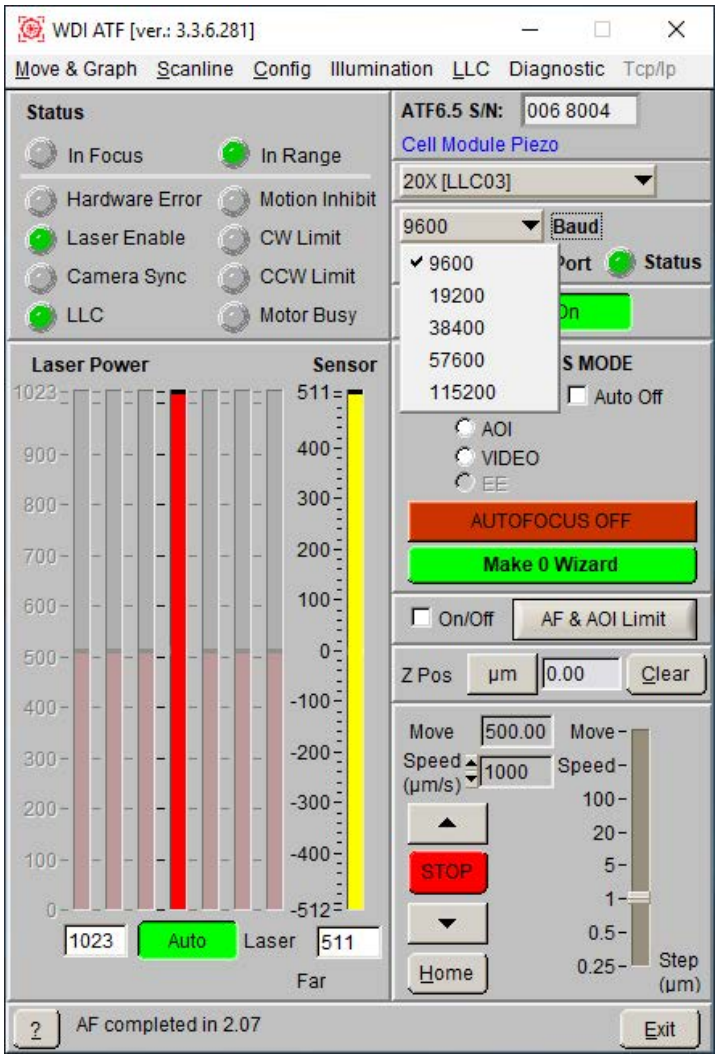
 © WDI Wise Device Inc.

Setting the initial sensor position

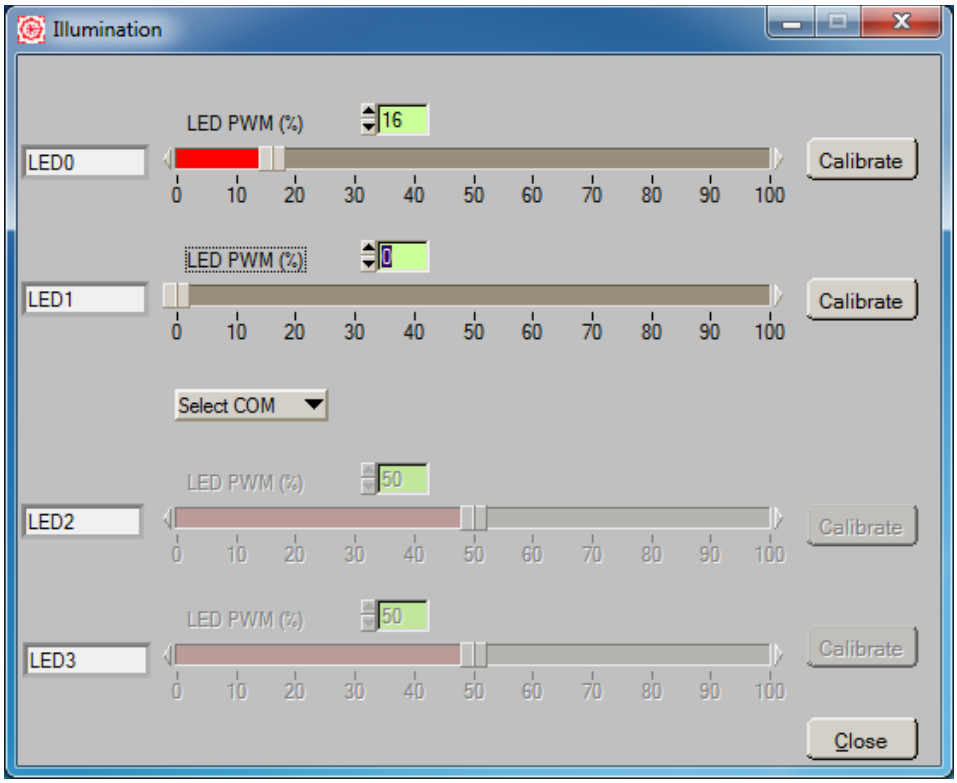
The first step in system alignment is to set the initial sensor position.

To set the initial sensor position:

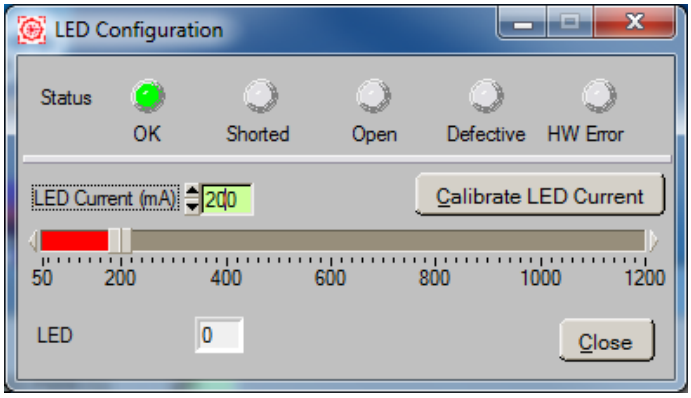
1. Install an objective lens with a low magnification (e.g., 5X). This will provide the best FOV for initial alignment.
2. Place the target sample under the microscope objective lens.
3. Start the **ATF Console** software, if it is not already running.
4. On the main panel, establish communication with the sensor, as follows:
   a. In the **Port** list, select the serial COM port where the sensor is connected.
   b. In the **Baud** list, select the required baud.



5. In the objective lens list, select the appropriate objective lens (e.g., 5X).
6. Check the camera image and ensure that the brightness and color quality of the image are appropriate for the sample. If necessary, adjust the illumination of the LED, as follows:
   a. On the **Illumination** menu, click **LED**.
   b. In the **Illumination** dialog box, next to the appropriate LED (normally **LED 0**), adjust the **LED PWM (%)** value to obtain an acceptable level of brightness on the camera image.
   c. To the right of the same LED, click **Calibrate**.



d. In the **LED Configuration** dialog box, adjust the **LED Current (mA)** value.



e. Click **Calibrate LED Current**. The LED should flash twice and then stay on.

f. Check the brightness and color quality of the camera image. If necessary, repeat steps b) to e) until the brightness and color quality are acceptable for your application.

g. Close the LED Configuration dialog box, and then close the Illumination dialog box.

7. Adjust the Z-stage position to obtain a sharp, in-focus image, as seen on the camera image. For details on how to move the Z-stage, see *"Adjusting the Z-stage position" on page 51*.

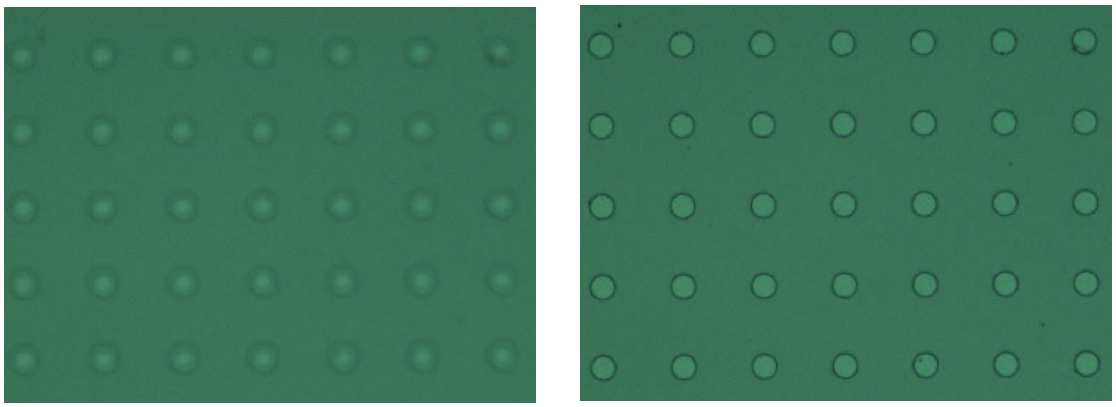


Figure 16 *Example of out-of-focus vs. in-focus image using 5X objective*

Verifying the objective parfocality

For optimal system performance, you must ensure that the focal points of the objectives lenses used in the system are in the same plane so that they can be interchanged without changing the focus.

To verify the objective parfocality:

1. Install an objective lens with a high magnification (e.g., 50X or higher).
2. If the target sample is not visible, adjust the illumination of the LED. For details, see *step 6 on page 58*.
3. Adjust the Z-stage position until the target sample is in the best focus position as seen on the camera image. For details, see *"Adjusting the Z-stage position" on page 51*.
4. Install an objective lens with a low magnification (e.g., 5X). Do not change the position of the sample or the Z-stage.
5. Adjust the height of the camera until the sample is in the best focus position.
6. Repeat steps 1 to 5 until there is little or no offset between the best focus position of the high magnification objective (50X) and that of the low magnification objective (5X).

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Centering the laser beam on the target

After setting the initial sensor position, you must ensure that the laser beam is centered on the target (i.e., positioned at the center of the camera FOV within ±30%).

NOTE: *The review camera must be aligned with the microscope's optical axis to be able to use it to perform this procedure.*

When the laser beam is centered on the target, the camera image appears as shown in *Figure 17*. If the laser beam is not centered on the target, use the following procedure to center the laser beam.

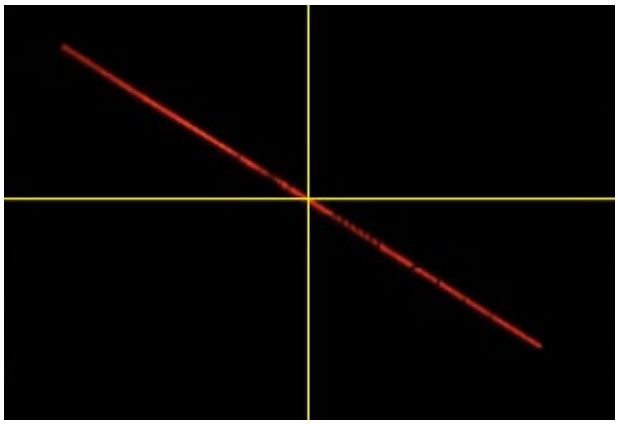
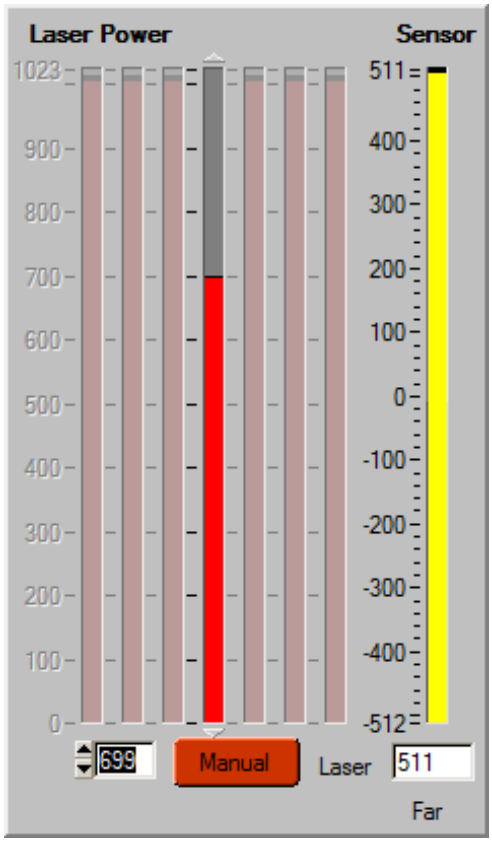


Figure 17 *Laser beam centered on target*

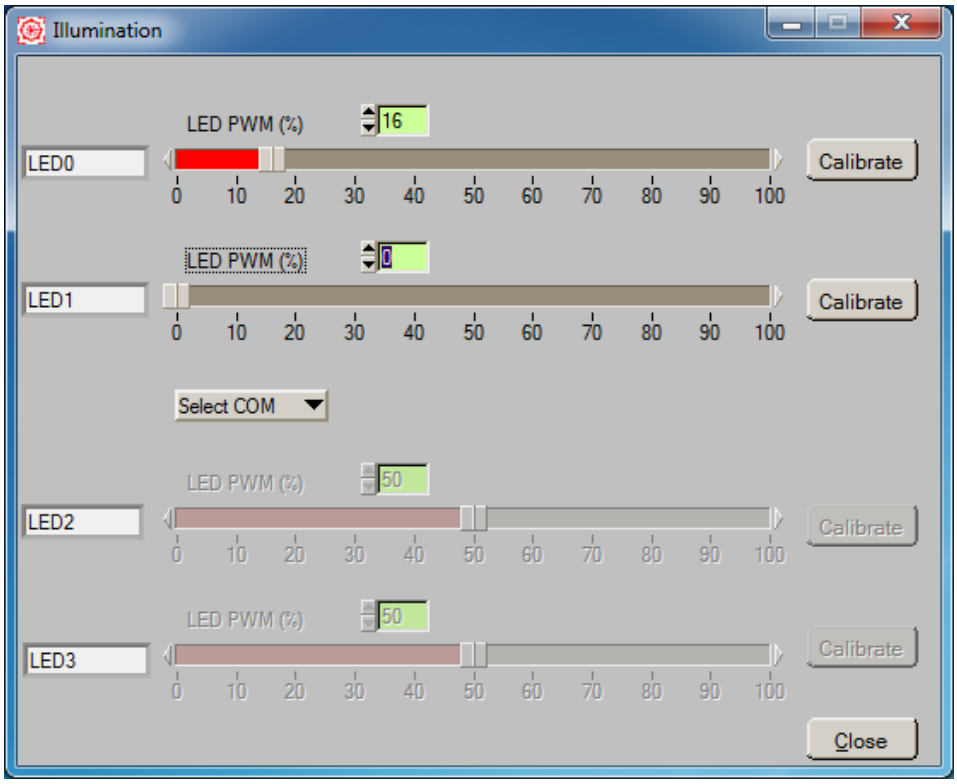
To center the laser beam on the target:

1. Ensure that the laser line is visible on the camera image.
2. If you need to improve the visibility of the laser beam, increase the laser power, as follows:
   a. On the main panel, under **Laser Power**, click **Auto**. The laser power changes to Manual mode.
   b. Drag the center bar to the required level, or type a value in the box and press ENTER.



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3. If the laser beam is still not clearly visible, reduce the illumination, as follows:
   a. On the **Illumination** menu, click **LED**.
   b. In the **Illumination** dialog box, next to **LED0**, reduce the **LED PWM (%)** value, and then click **Close**.



4. Ensure that the laser beam is positioned at the center of the camera FOV within ±30%. If it is not, check the alignment of the optical components of the system.
5. When you are finished, return the **Laser Power** to **Auto** mode and increase the illumination, if necessary.

Centering the laser beam on the sensor

After centering the laser beam on the target, you must ensure that the return laser beam is centered on the ATF 6.5 sensor detector.

The sensor has two adjustment screws (painted green) that are used to adjust the alignment of the laser beam with the following effects:

- **Adjuster A** – Shifts the horizontal position of the magenta peak in the Scanline window.
- **Adjuster B** - Shifts the horizontal position of the green peak in the Scanline window.

You should not have to turn the adjusters more than one full turn in any direction.

CAUTION: Do not adjust the screws on the sensor that are painted red.

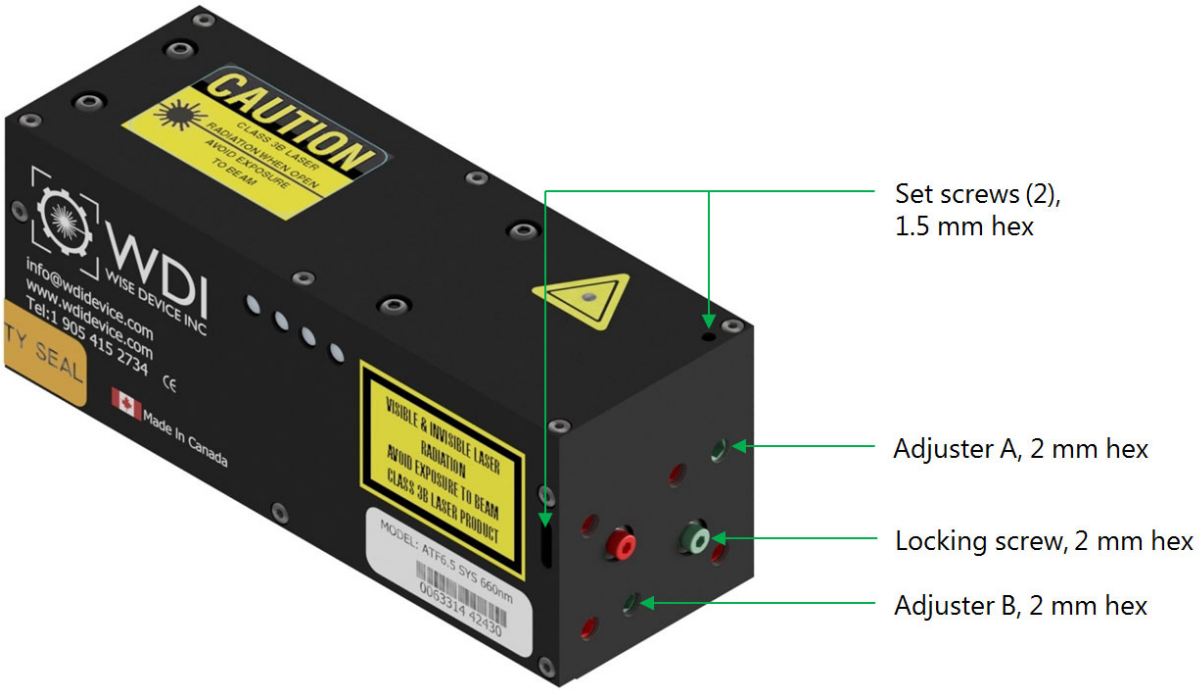
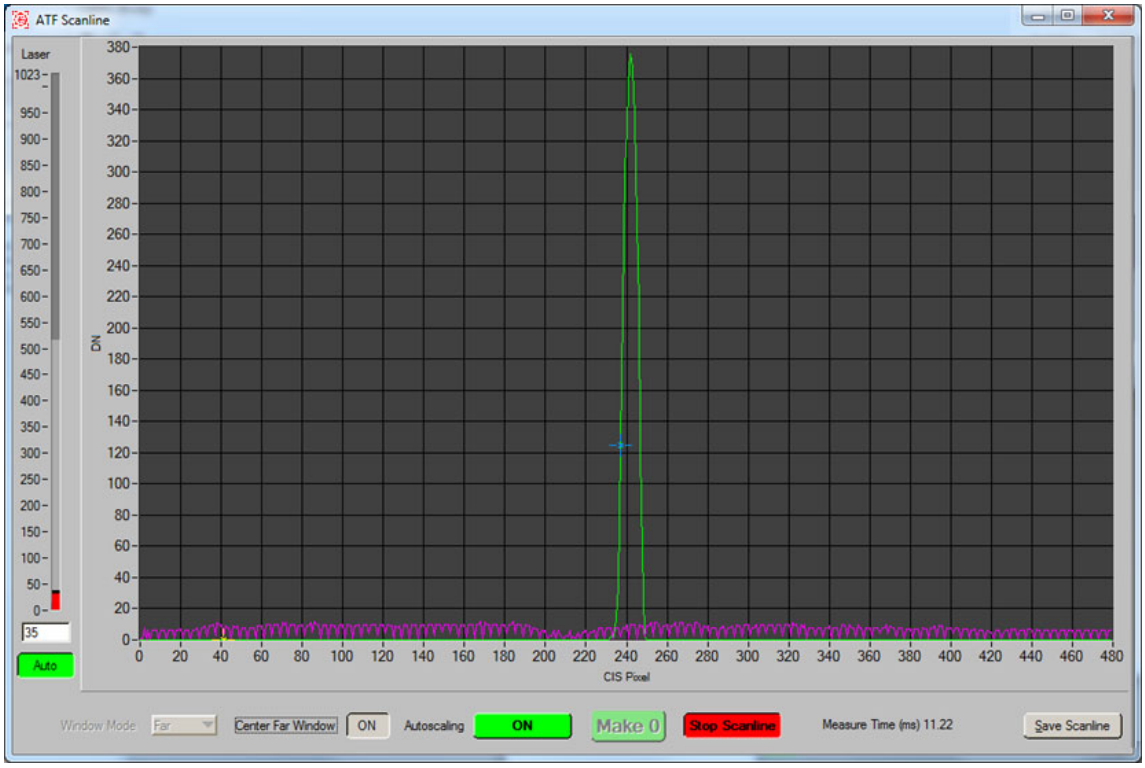


Figure 18 *Sensor adjustment screws*

To center the laser beam on the sensor:

1. On the main menu, click **Scanline**.
2. At the bottom of the **ATF Scanline** window, in the **Window Mode** list, click **Far**. Using Far mode will help to ensure that the peaks remain centered when different objective lenses are selected.
3. Next to **Center Far Window**, click **OFF**. The button changes to ON.
4. Check the position of the green and magenta peaks. The green peak should be tall and narrow at the lowest effective laser power, and both peaks should be centered at 240 ±3 pixels. The amplitude of the magenta peak is very small, so you might need to increase the laser power slightly to better visualize the peak of the magenta curve. To increase the laser power, drag the slider on the **Laser** bar upward slightly until you can see the peak.



5. If the peaks are not centered at 240 pixels, adjust the position of the peaks using the sensor adjustment screws (see *Figure 18*):
    a. Release the set screws that hold the two adjustment screws in place during shipping by turning them one half turn CCW. You do not need to re-tighten the set screws after adjustment unless the sensor will be used in a high vibration application (e.g., mounted in a gantry).
    b. Release the locking screw by turning it one turn CCW.
    c. To adjust the horizontal position of the magenta peak, turn **Adjuster A** very slowly while observing the Scanline.
    d. To adjust the horizontal position of the green peak, turn **Adjuster B** very slowly while observing the Scanline.
    e. When both peaks are centered at 240 ±3 pixels, tighten the locking screw on the sensor. Be careful not to over-tighten the locking screw.
6. Close the **ATF Scanline** window. Window Mode automatically resets to Auto when you close the window.

Aligning the half aperture for low magnification

In this procedure, the Move & Graph window is used to validate the sensor output by calculating the slope of the line, which plots sensor output versus Z-stage motor position, and comparing it to the system's default slope parameter. For best results, the calculated slope should be within ±20% of the default slope parameter. If the calculated slope is not within this range, you must adjust the mechanical position of the sensor within the slotted screw holes on the mounting bracket.

NOTE: *The initial default slope parameter (i.e., from the factory settings) is for a Mitutoyo M Plan Apo objective. This value might not be accurate for a different type of objective lens. If you have a different type of objective lens and are having issues meeting the requirements of this procedure, please contact our Applications Team at applications@wdidevice.com.*

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The half aperture in the title refers to the laser light shape that is viewed on one half of the objective's entry aperture, which resembles a rectangle, as shown in *Figure 19*.

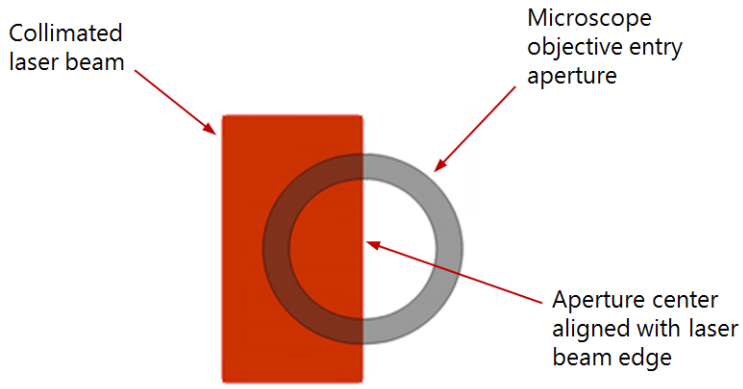
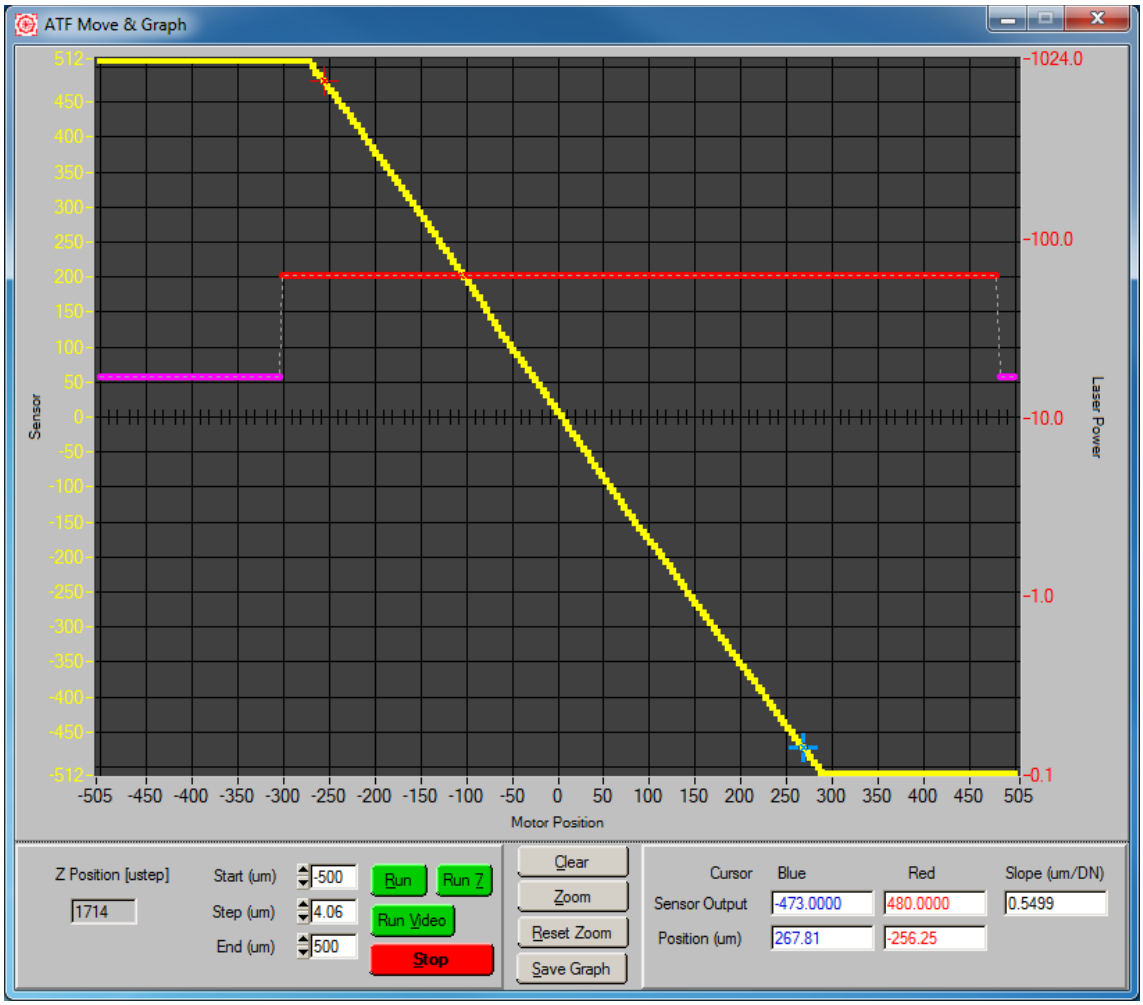


Figure 19 *Half aperture alignment*

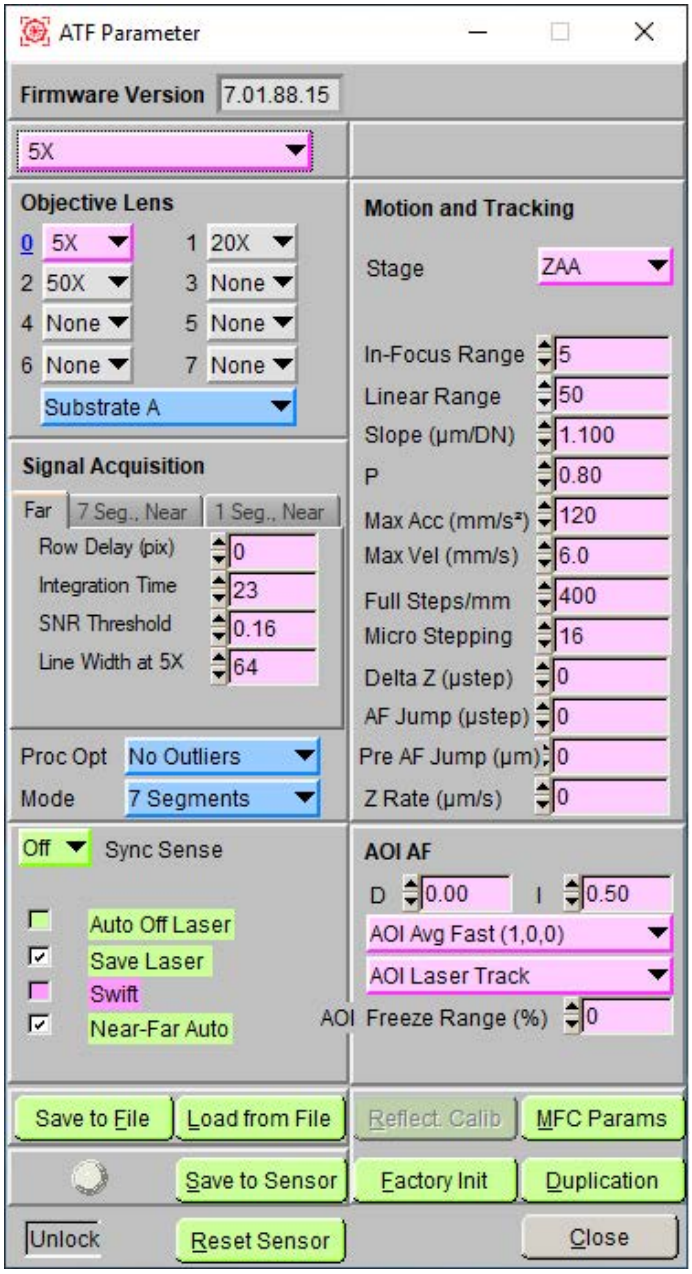
To align the sensor half aperture for a low magnification objective lens:

1. Install an objective lens with a low magnification (e.g., 5X).
2. Ensure that the target sample is in focus, as seen on the camera image. If it is not, adjust the Z-stage position to obtain a sharp, in-focus image. For details, see *"Adjusting the Z-stage position" on page 51*.
3. To set the current focus position as the zero position, perform manual Make 0. For details, see *"Performing manual Make 0" on page 53*.
4. On the main menu, click **Scanline**.
5. At the bottom of the **ATF Scanline** window, in the **Window Mode** list, click **Far**.
6. Next to **Center Far Window**, click **OFF**. The button changes to ON.
7. Ensure that the green peak is tall and narrow with the laser power at its lowest level, and both the green and magenta peaks are centered at 240 ± 3 pixels. If they are not, adjust the sensor as described in *"Centering the laser beam on the sensor" on page 64*.
8. On the main menu, click **Move & Graph**.
9. At the bottom of the **ATF Move & Graph** window, enter the following values:
   - In the **Start (µm)** box, type **-500**.
   - In the **Step (µm)** box, type **4**.
   - In the **End (µm)** box, type **500**.
10. Click **Run**.

11. When the graph is complete, calculate the slope of the yellow line at its most linear part:
    a. Click near the bottom of the yellow diagonal line. A blue cursor (+) is displayed.
    b. Click near the top of the yellow diagonal line. A red cursor (+) is displayed.
    c. The system calculates the slope of the line between the two cursors automatically and displays it in the **Slope (µm/DN)** box.
    d. Record the value.



12. Find the default slope value, as follows:
    a. In the main panel, on the **Config** menu, click **Parameters**.
    b. Under **Motion and Tracking**, view the default value in the **Slope (µm/DN)** box.
    c. Record the value.



13. Compare the slope value calculated in the Move & Graph window to the default slope value in the Parameter dialog box. For best results, the calculated slope should be within ±20% of the default slope. If it is not, adjust the position of the sensor, as follows:
    - If the calculated slope value is *more than* 20% of the default slope value, loosen the mounting screws and move the sensor down approximately 0.25 mm.
    - If the calculated slope value is *less than* 20% of the default slope parameter, loosen the mounting screws and move the sensor up approximately 0.25 mm.

    For details, refer to *Figure 7* in *"Mounting bracket" on page 27*. You might have to adjust the sensor position several times to find the best position.

Aligning the half aperture for high magnification

This procedure uses a high magnification objective lens to perform a more precise version of the procedure described in *"Aligning the half aperture for low magnification" on page 66*.

To align the sensor half aperture for a high magnification objective lens:

1. Install an objective lens with a high magnification (e.g., 50X).
2. Ensure that the target sample is in focus, as seen on the camera image. If it is not, adjust the Z-stage position to obtain a sharp, in-focus image. For details, see *"Adjusting the Z-stage position" on page 51*.
3. To set the current focus position as the zero position, perform manual Make 0. For details, see *"Performing manual Make 0" on page 53*.
4. On the main menu, click **Scanline**.
5. In the **ATF Scanline** window, ensure that the green peak is tall and narrow with the laser power at its lowest level, and both the green and magenta peaks are centered. If they are not, adjust the sensor as described in *"Centering the laser beam on the sensor" on page 64*.
6. On the main menu, click **Move & Graph**.
7. Click **Run**.
8. Check the peaks on the Scanline window to see if the sensor is in the correct position:
    a. On the main menu, click **Scanline**.
    b. At the bottom of the **ATF Scanline** window, in the **Window Mode** list, click **Far**.
    c. Next to **Center Far Window**, click **OFF**. The button changes to ON.

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d. When in focus, the green peak should be centered at 240 ±3 pixels, as shown by the **In focus** image of *Figure 20*.

e. Move the Z-stage up 20 µm. The green peak should move to the left, and the right side of the peak should begin below 240 pixels.

f. Move the Z-stage down 20 µm. The green peak should move to the right, and the left side of the peak should begin above 240 pixels.

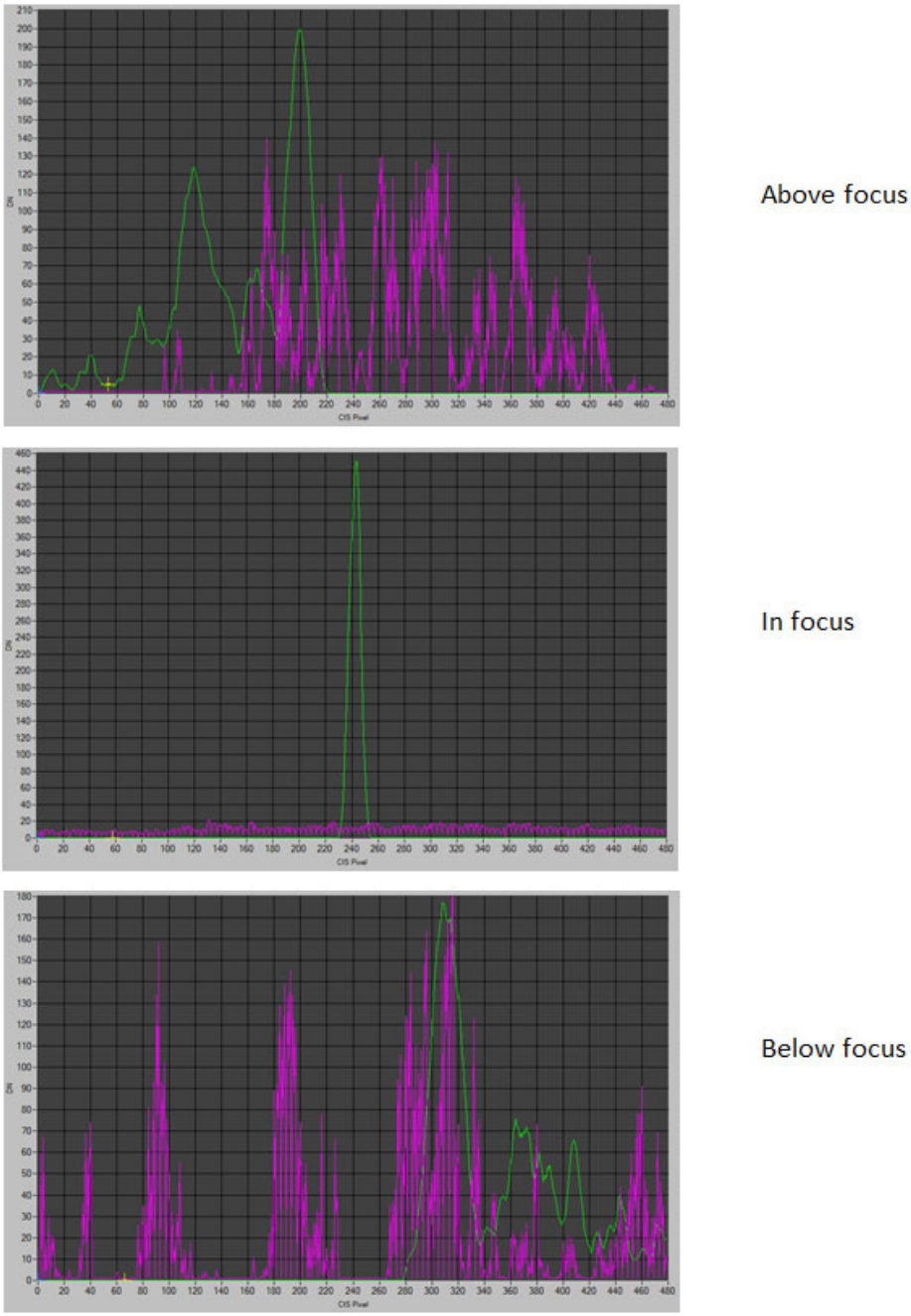


Figure 20 *Scanline peaks on well-aligned sensor with 50X objective*

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CHAPTER

5

Pairing the sensor with the objectives

This chapter describes how to pair the sensor with each objective lens used in the system.

The following topics are covered:

- *Pairing overview, pg. 74*
- *Pairing the sensor with an objective lens, pg. 74*
- *Saving the parameters, pg. 78*

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Pairing overview

You must pair the sensor with each objective lens used in the system and for each different type of sample. Pairing must be performed immediately after aligning the system (see *"Aligning the system" on page 55*), as well as each time a new objective lens or type of sample is introduced to the system.

NOTE: *The procedures in this chapter assume that the ATF Console software has been set up as described in "Setting up the ATF Console software" on page 46, and that the system has been aligned as described in "Aligning the system" on page 55.*

At the same time that you pair the sensor with the objectives, you can update the objective-dependent parameters. For autofocus to work optimally for the selected objective lens, the Slope parameter should be calculated and the value entered in the Parameter dialog box.

NOTE: *The initial default slope parameter (i.e., from the factory settings) is for a Mitutoyo M Plan Apo objective. This value might not be accurate for a different type of objective lens. If you have a different type of objective lens and are having issues meeting the requirements of this procedure, please contact our Applications Team at applications@wdidevice.com.*

Pairing the sensor with an objective lens

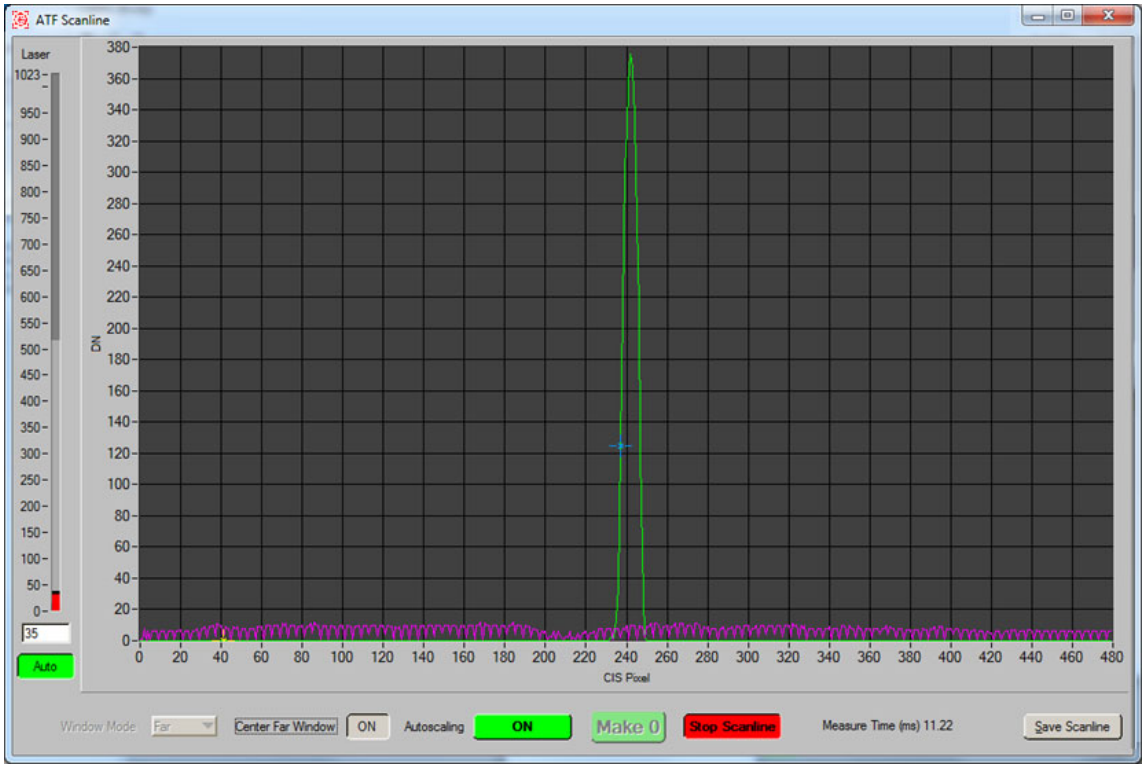
You must perform this procedure for each objective lens used in your system and for each different type of sample.

To check the sensor configuration for an objective lens:

1. Start the **ATF Console** software, if it is not already started.
2. On the main panel, in the **Baud** list, select the required baud.
3. In the **Port** list, select the serial COM port where the sensor is connected.
4. In the objective lens list, select the objective lens that you want to pair with the sensor.

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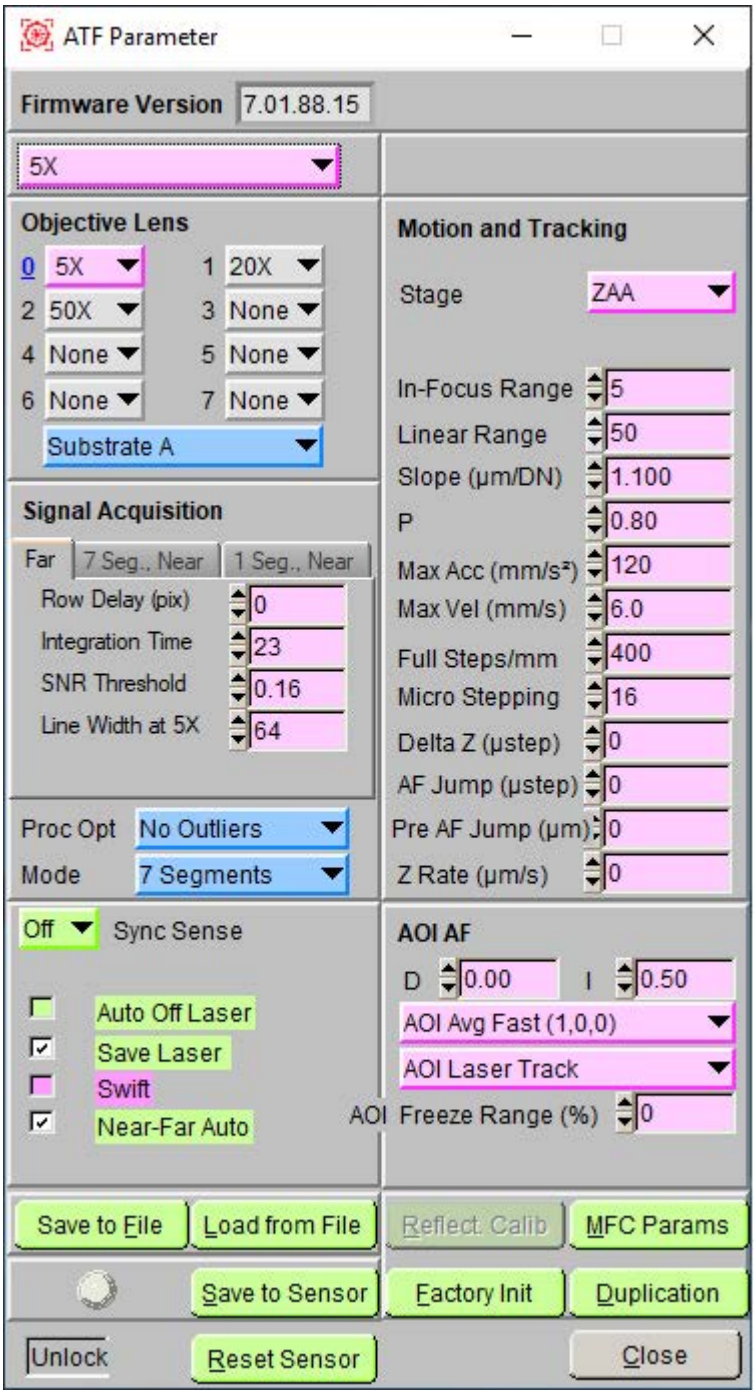
5. Ensure that the sample is in focus, as seen on the camera image. If it is not, adjust the Z-stage position to obtain a sharp, in-focus image. For details, see *"Adjusting the Z-stage position" on page 51*.
6. To set the current focus position as the zero position, perform manual Make 0. For details, see *"Performing manual Make 0" on page 53*.
7. On the main menu, click **Scanline**.
8. At the bottom of the **ATF Scanline** window, in the **Window Mode** list, click **Far**.
9. Next to **Center Far Window**, click **OFF**. The button changes to ON.
10. Ensure that the green peak is tall and narrow with the laser power at its lowest level, and both the green and magenta peaks are centered at 240 ±3 pixels. If the peaks are not centered, adjust the sensor as described in *"Centering the laser beam on the target" on page 61*.



11. On the main menu, click **Move & Graph**.
12. At the bottom of the **ATF Move & Graph** window, enter the required values in the **Start (µm)**, **Step (µm)**, and **End (µm)** boxes, depending on the linear range of the selected objective lens.
13. Click **Run**.
14. When the graph is complete, calculate the slope of the yellow line at its most linear part:
    a. Click near the bottom of the yellow diagonal line. A blue cursor (+) is displayed.
    b. Click near the top of the yellow diagonal line. A red cursor (+) is displayed.
    c. The system calculates the slope of the line between the two cursors automatically and displays it in the **Slope (µm/DN)** box.
    d. Record the value.

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15. Find the default slope value, as follows:
    a. In the main panel, on the **Config** menu, click **Parameters**.
    b. Under **Motion and Tracking**, view the default value in the **Slope (µm/DN)** box.
    c. Record the value.



16. Compare the slope value calculated in the Move & Graph window to the default slope value in the Parameter dialog box. For best results, the calculated slope should be within ±20% of the default slope.
    - If the calculated slope value is within the acceptable range, in the **Parameters** dialog box, enter the calculated value in the **Slope (µm/DN)** box. This will ensure that autofocus works optimally for the selected objective lens.
    - If the calculated slope value is not within the acceptable range, adjust the sensor mechanical position using the mounting screws shown in *Figure 7*, as follows:
        - If the calculated slope value is *more than* 20% of the default slope value, loosen the mounting screws and move the sensor down approximately 0.25 mm.
        - If the calculated slope value is *less than* 20% of the default slope parameter, loosen the mounting screws and move the sensor up approximately 0.25 mm.

Saving the parameters

When you are finished checking the sensor configuration for all the objective lenses, it is good practice to save the updated parameters to the sensor. When you save the parameters to the sensor, the settings become the sensor default settings that are restored when you click **Reset Sensor**.

To save the sensor parameters:

1. When you are finished updating the parameters for all the objective lenses, at the bottom of the **Parameters** dialog box, click **Save to Sensor**.
2. Click **Close** to close the **Parameter** dialog box.

Saving the laser settings

When you save the settings to the laser, the settings become the sensor's default laser settings.

To save the laser settings:

1. When you are finished updating the parameters for all the objective lenses, at the bottom of the **Parameters** dialog box, click **Save Laser**.
2. Click **Close** to close the **Parameter** dialog box.

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CHAPTER

6

Configuring CM

This chapter describes how to configure the CM feature.

The following topics are covered:

- *CM configuration overview, pg. 80*
  - *Surface validation, pg. 80*
- *Configuring a CM sample, pg. 84*
  - *Configuring CM using a 5X objective lens, pg. 84*
  - *Configuring CM using a 50X objective lens, pg. 90*
- *Verifying CM configuration, pg. 93*

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CM configuration overview

The CM (cell and module) feature analyzes and differentiates the reflections from each surface of a multi-layer medium, such as an LCD panel, to focus on a selected surface.

This chapter describes, in general, how to configure a CM sample; however, keep in mind that samples can vary greatly and you must adjust the CM parameters to suit your specific application.

Surface validation

The CM feature uses surface validation to identify the correct surface for focus convergence. To understand how surface validation works, we first need to define the top, bottom, and stage surfaces as they relate to the CM sample. As an example, refer to *Figure 21*.

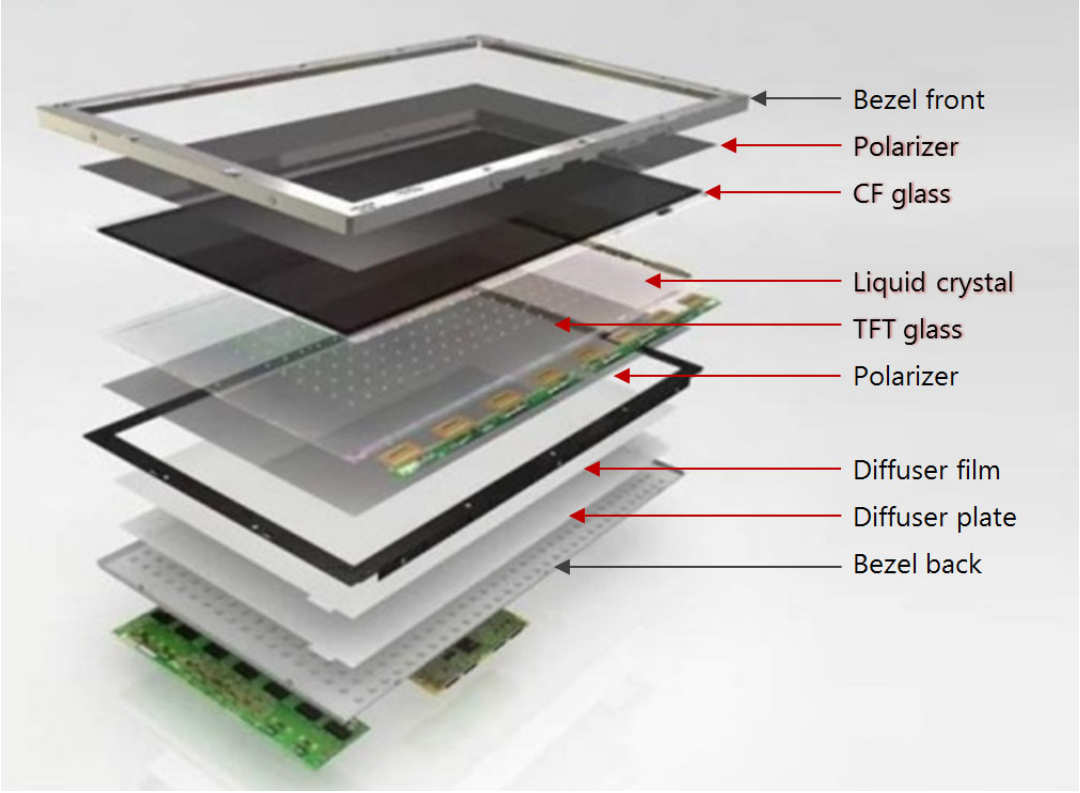


Figure 21 *CM sample surfaces*

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The surfaces are defined as follows:

- **Top surface** – The top-most glass surface of the CM sample. This is the first surface the system will focus on while approaching the sample from above. It is normally the surface with the strongest reflection, such as CF glass. If a polarizer is used, the top surface is typically the top of the polarizer.
- **Bottom surface** – The bottom glass surface of the CM sample. This is the second surface the system will focus on while approaching the sample from above. Depending on the sample, this can be the liquid crystal layer or TFT glass.
- **Stage surface** – The bottom-most surface of the CM sample that sits on the stage. This is the third surface the system will focus on while approaching the sample from above. In typical applications, the stage surface is weak and difficult to focus on since there are multiple surfaces above it. High NA lenses might differentiate between the TFT and CF sides, making the stage surface the fourth surface.

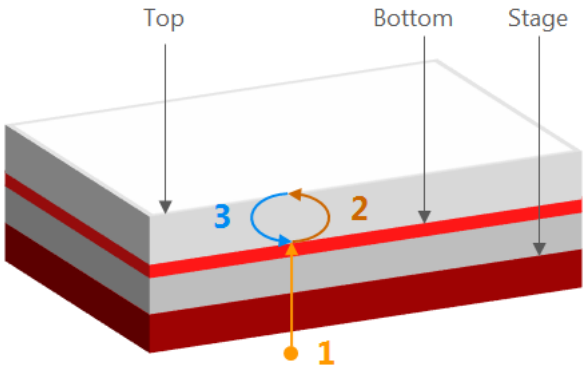
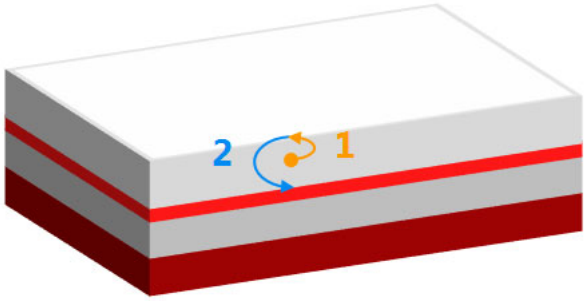
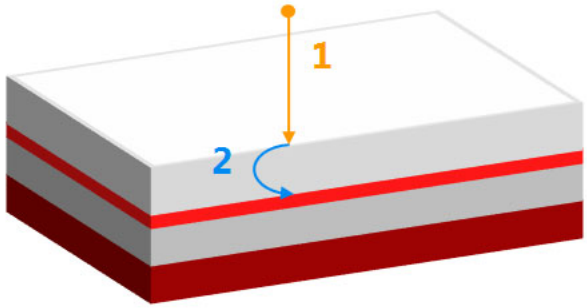
For objectives greater than 10X, surface validation is able to unambiguously distinguish the top surface from the bottom surface only if the current surface is the top surface. For this reason, the surface validation algorithm executes a stage motion jump, equivalent to the optical glass thickness, upward until the surface is identified as the top surface. At this point, if the requested surface is the bottom surface (as in bottom validation), the stage is directed to execute an additional motion down. In practical terms, this means that if the system starts focusing when it is close to the bottom surface, it will execute two additional steps, a quick jump up and then another quick jump back down, to reach the bottom surface. While this approach adds about a second to the focusing time, it assures that the system reaches the correct destination surface.

There are two types of surface validation:

- **Bottom validation** – Validates the in-focus position on the bottom surface.
- **Top validation** – Validates the in-focus position on the top surface.

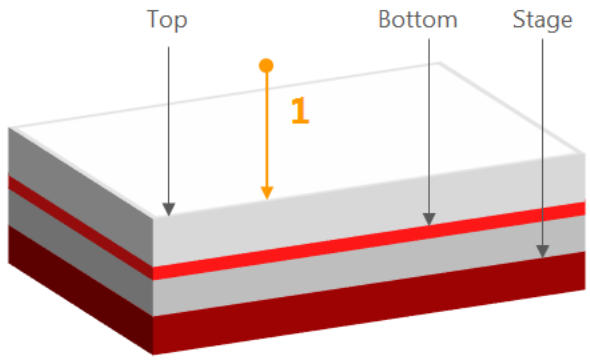
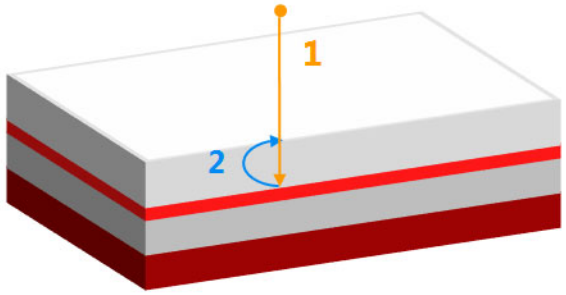
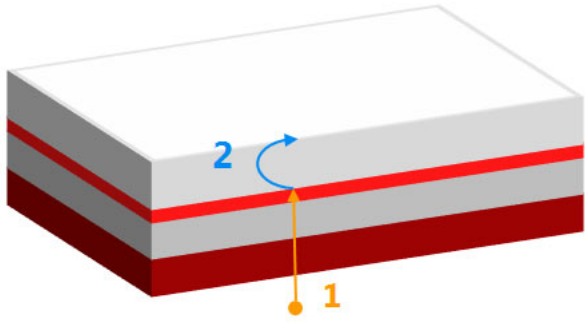
We recommend using bottom validation for samples that have multiple layers. Top validation is normally used only for samples that have one layer. *Table 12* illustrates the algorithm for bottom validation, and *Table 13* illustrates the algorithm for top validation.

Table 12 Bottom validation

	Example #1 – The current focus point is located below the bottom surface of the CM sample. The process is: 1. Focus on the bottom surface. 2. Jump to the top surface and validate. 3. Jump to the bottom surface and into bottom surface in-focus position.
	Example #2 – The current focus point is located between the top and bottom surfaces (but closer to the top surface). The process is: 1. Focus on the top surface. 2. Jump to the bottom surface and into the bottom surface in-focus position.
	Example #3 – The current focus point is located above the top surface of the CM sample. The process is: 1. Focus on the top surface. 2. Jump to the bottom surface and into the bottom surface in-focus position.

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Table 13 Top validation

	Example #1 – The current focus point is located above the top surface of the CM sample. The process is: 1. Focus in one move to until reach the top surface in-focus position
	Example #2 – The current focus point is located between the top and bottom surfaces (but closer to the bottom surface). The process is: 1. Focus on the bottom surface. 2. Jump to the top surface and into the top surface in-focus position.
	Example #3 – The current focus point is located below the bottom surface of the CM sample. The process is: 1. Focus on the bottom surface. 2. Jump to the top surface and into the top surface in-focus position.

Configuring a CM sample

Because surface validation is sample-dependent, you must configure a sample so that the review camera can recognize the different sample surfaces. In this section, a CF sample is configured using bottom validation, which is the recommended surface validation type.

The procedures in this section assume that the system has already been configured for standard operation, meaning that autofocus is working as intended on regular (not TFT or CF) substrates.

NOTE: *If your sample has more than three surfaces, make the desired surface the bottom surface; thus, the surface above the bottom surface will be the top surface, and the surface below the bottom surface will be the stage surface. For most samples, begin with bottom surface validation.*

When configuring a CM sample, you must perform Make 0. Here are some useful tips for performing Make 0 for CM samples:

- Make 0 must be performed on the desired surface (Top or Bottom).
- The blue matrix on the CF panel side makes a good target.
- On CF panels, you can put all the spots on either the blue matrix or the green pixels. Avoid the red pixels because they generate large bottom reflections that can smear the peak position.
- On TFT panels, the pads close to the panel edge provide enough clear glass or metal covered surface for all the spots.
- On TFT panels, be aware that the pixel area contains patterns (ITO fingers) that may shift the spot position and unnecessarily expand the peak widths. Place all the spots on metal, or look at the edge of the panel for clear and undisturbed sections instead.

Configuring CM using a 5X objective lens

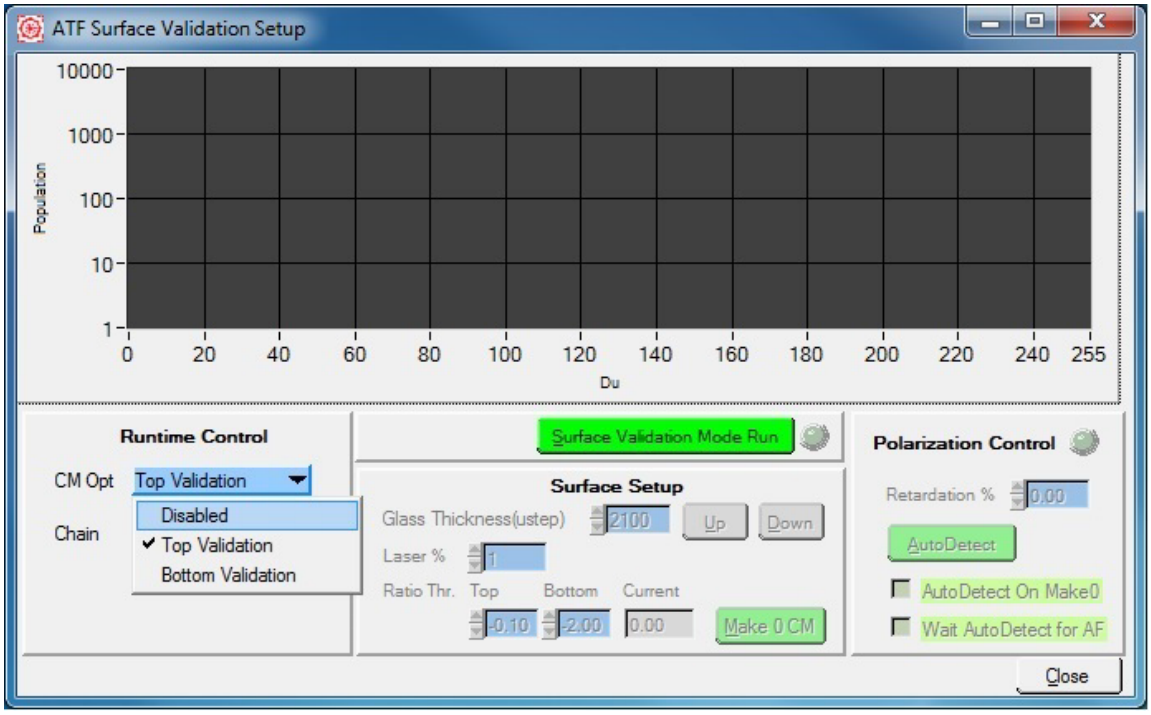
This procedure describes how to set up and run the sensor on a CF sample using a 5X objective lens. It is important to verify that the 5X objective lens is operating normally before attempting to configure the 50X objective lens.

Before you begin, ensure that the sensor is optically aligned with a 50/50 beamsplitter and that the ATF Console software is connected to the sensor with the highest speed for ease of operation.

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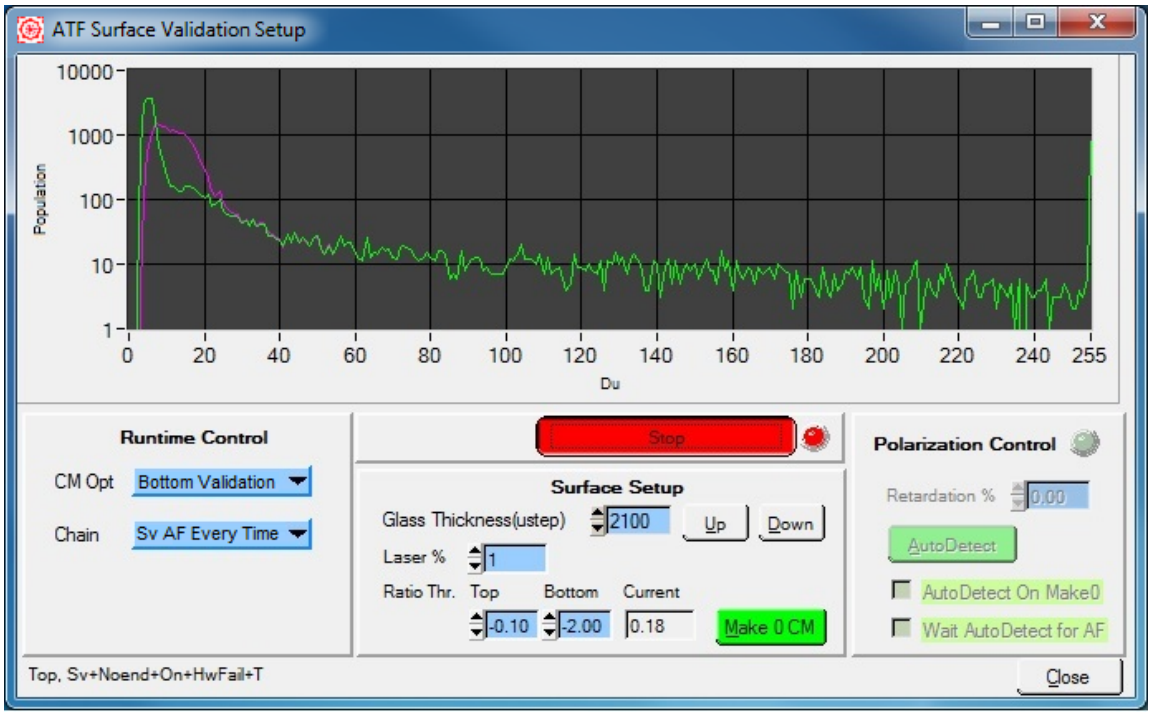
To configure a 5X objective lens:

1. Install a 5X objective lens.
2. Bring the sample into focus on the bottom surface.
3. Adjust the illumination for clear visibility of the sample, as follows:
   a. On the **Illumination** menu, click **LED**.
   b. Next to **LED1**, in the **LED PWM (%)** box, adjust the value, and then click **Close**.
4. To suppress the influence of the strong background reflection from the bottom surface while in focus, use the **Filter Peaks** processing option:
   a. On the **Config** menu, click **Parameter**.
   b. Under **Signal Acquisition**, in the **Proc Opt** list, click **Filter Peaks**.
5. On the **Config** menu, click **CM**.
6. In the **ATF Surface Validation Setup** window, in the **CM Opt** list, click **Bottom Validation**.



7. Perform manual Make 0. For details, see *"Performing manual Make 0" on page 53*.

8. If you do not know the glass thickness (i.e., the distance between the top surface and the bottom surface), measure the glass thickness, as follows:
   a. Focus on the bottom surface and adjust the Z-stage position to bring the reading on the **Sensor** bar graph on the ATF Console main panel to 0.
   b. Next to **AbsPos**, click **Clear**.
   c. Move to the top surface and adjust the Z-stage position to bring the reading on the **Sensor** bar graph to 0.
   d. Record the **AbsPos** value (in µm).
   e. Repeat the previous steps on a few areas of the sample. The glass thickness within a single sample can vary by as much as 30 µm. Take the maximum **AbsPos** value of all the recorded values and add 5%. Record this value.
9. Click **Surface Validation Mode Run.**
10. While the validation is running, do the following:
    a. In the **Glass Thickness** box, enter the value calculated in *step 8*.
    b. In the **Laser%** box, enter 1 (or another small value such as 2 or 3).



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11. Click **Make 0 CM**, which calculates and updates the **Ratio Thr.** values.

    Alternatively, you can perform this step manually by following steps a to f, below, which will help you understand how the Ratio Threshold values are calculated. The goal of CM configuration is to determine the Ratio Threshold (i.e., balance of light intensity from above and below) for each of the surfaces that the system needs to differentiate. CM can work correctly only when the Ratio Thr. values do not overlap. In addition, the laser power is critical and needs to be adjusted carefully (see previous step).

    a. Return to the main panel and perform manual **Make 0**. This time, Make 0 takes the correct glass thickness into account.

    b. While focusing on the bottom surface, under **Surface Setup**, read and record the value in the **Current** box. This value will be called Current Bottom.

    c. Click **Up** to jump and focus on the top surface.

    d. While focusing on the top surface, read and record the value in the **Current** box. This value will be called Current Top.

    e. Click **Down** to return to the bottom surface.

    f. Enter the **Ratio Thr.** values, as follows:

    – In the **Top** box, enter a value equal to (Current Bottom + Current Top)/2. This threshold allows the system to determine if focus is taking place on the top or bottom surface.

    – In the **Bottom** box, enter a value equal to Current Bottom. This threshold allows the system to differentiate between the stage and bottom surfaces.

12. On the main menu, click **Move & Graph**.

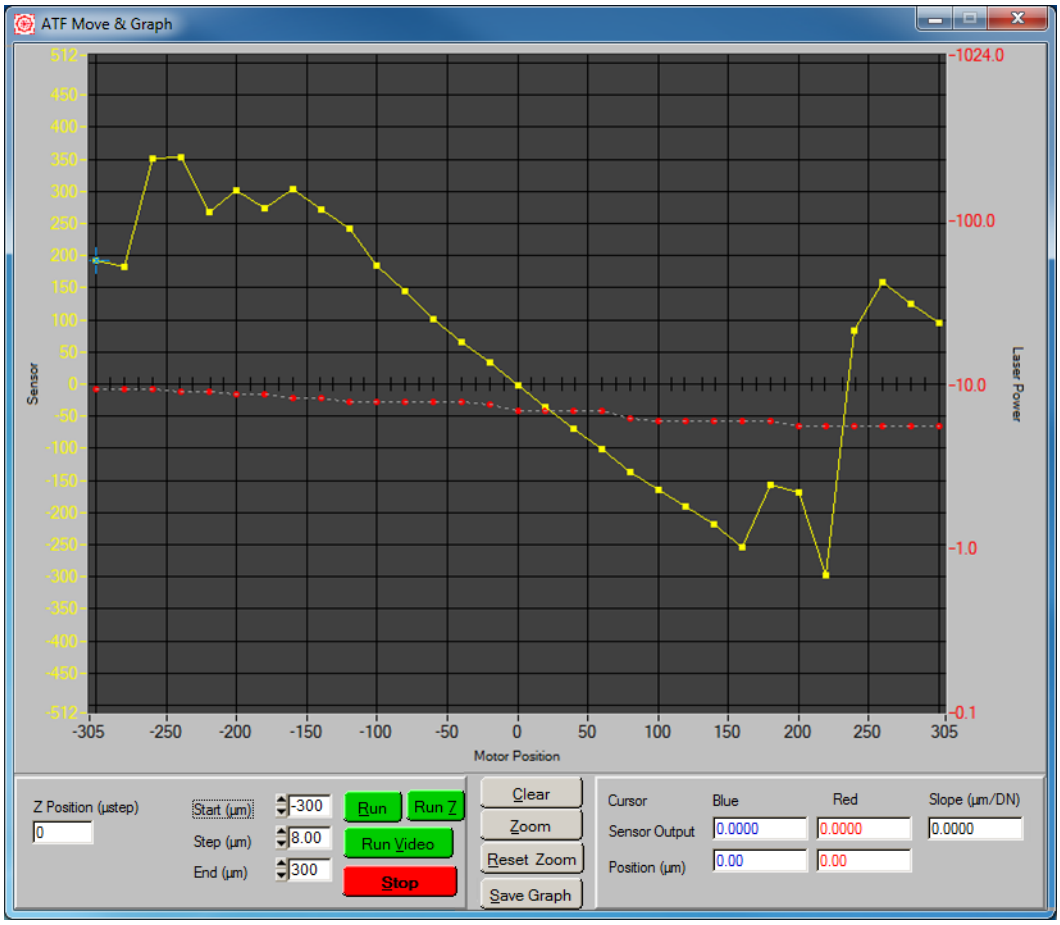
13. At the bottom of the **ATF Move & Graph** window, enter the following values:

    – In the **Start (µm)** box, type **-300**.

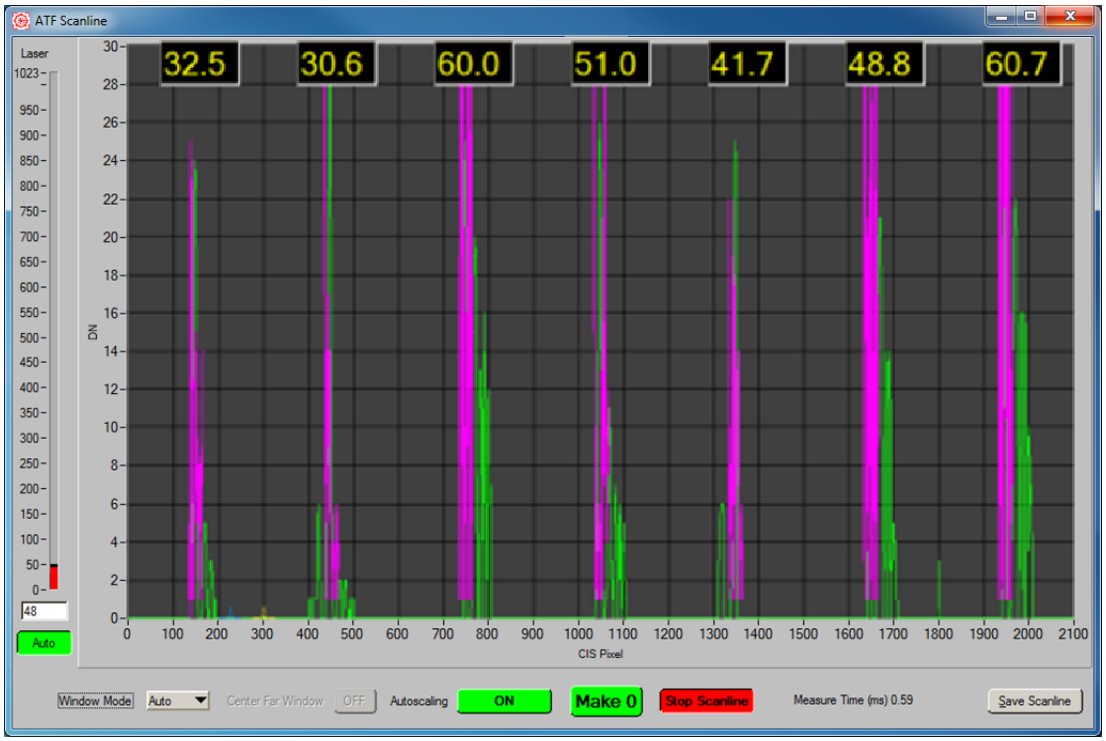
    – In the **Step (µm)** box, type **8**.

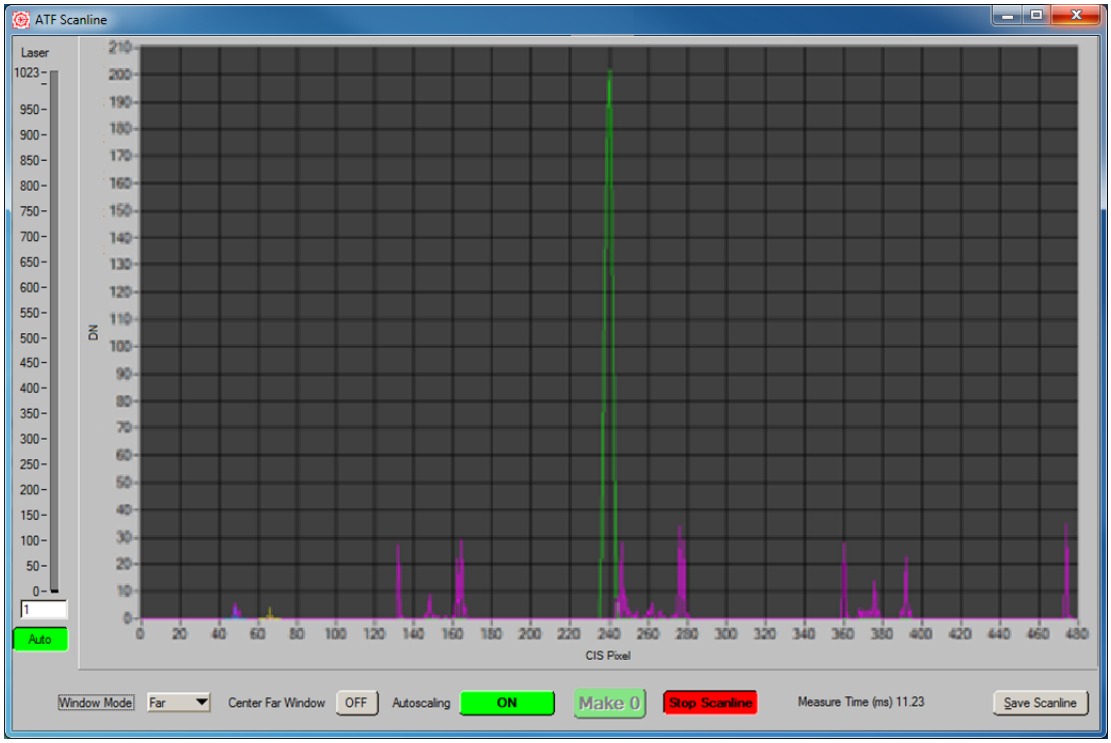
    – In the **End (µm)** box, type **300**.

14. Click **Run**. The resulting graph should have a linear section passing through zero and a varied signal above and below zero when the position is transitioning between the top and bottom surfaces. The important span of the linear signal should extend ±100 µm from the zero position. If it does not, the sensor or camera alignment should be verified.



15. To check the peaks, on the main menu, click **Scanline**. The Scanline images should appear as shown below for **Auto** mode and **Far** mode.



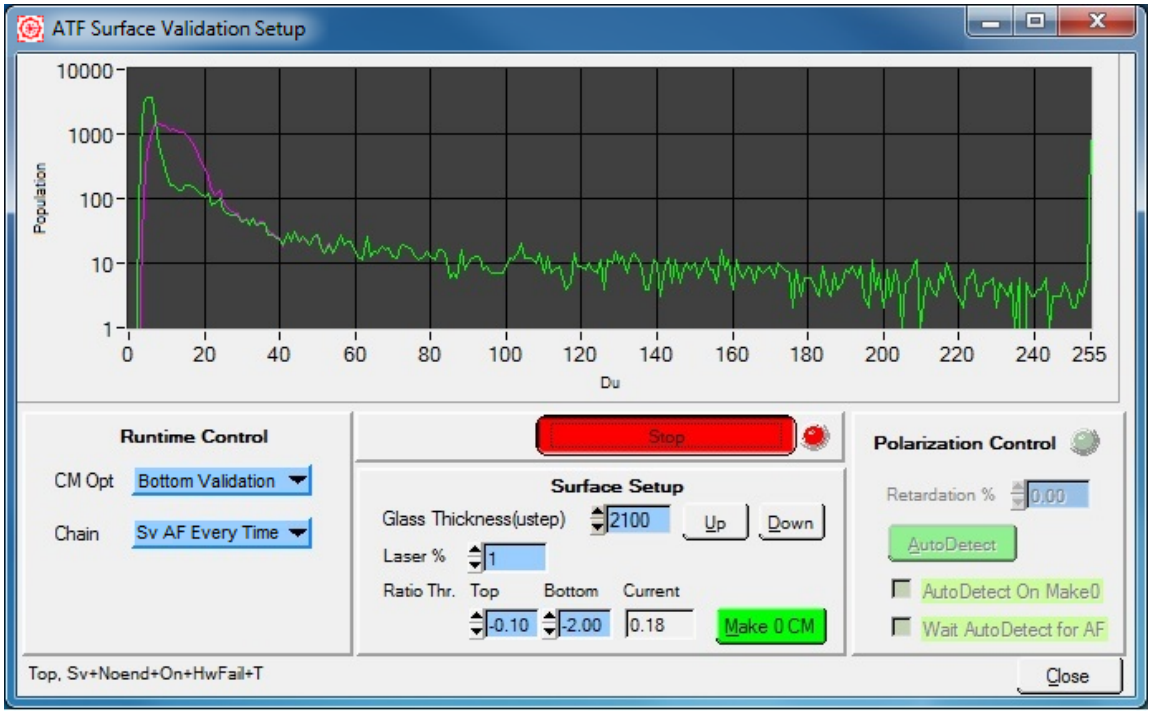


Configuring CM using a 50X objective lens

This procedure describes how to set up and run the sensor on a CF sample using a 50X objective lens.

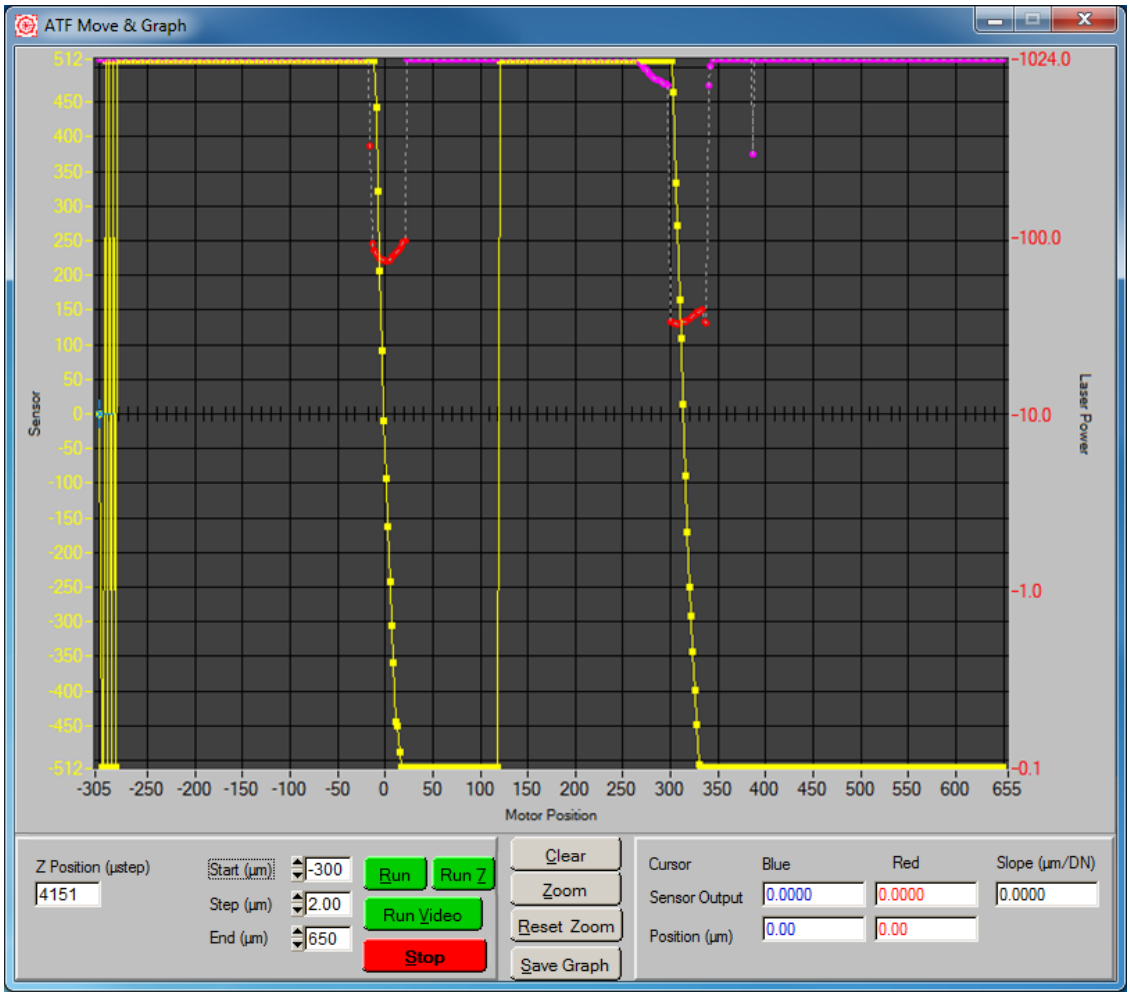
To configure a 50X objective lens:

1. Install a 50X objective lens.
2. Bring the sample into focus on the bottom surface.
3. Adjust the illumination for clear visibility of the sample on the camera, as follows:
   a. On the **Illumination** menu, click **LED**.
   b. Next to **LED1**, in the **LED PWM (%)** box, adjust the value, and then click **Close**.
4. On the **Config** menu, click **CM**.
5. In the **Surface Validation Setup** window, in the **CM Opt** list, ensure that **Bottom Validation** is selected.
6. Click **Surface Validation Mode Run**.
7. While the validation is running, in the **Laser%** box, type **70**. The 50X objective requires more laser power to distinguish between the top and bottom surfaces compared to the 5X lens.

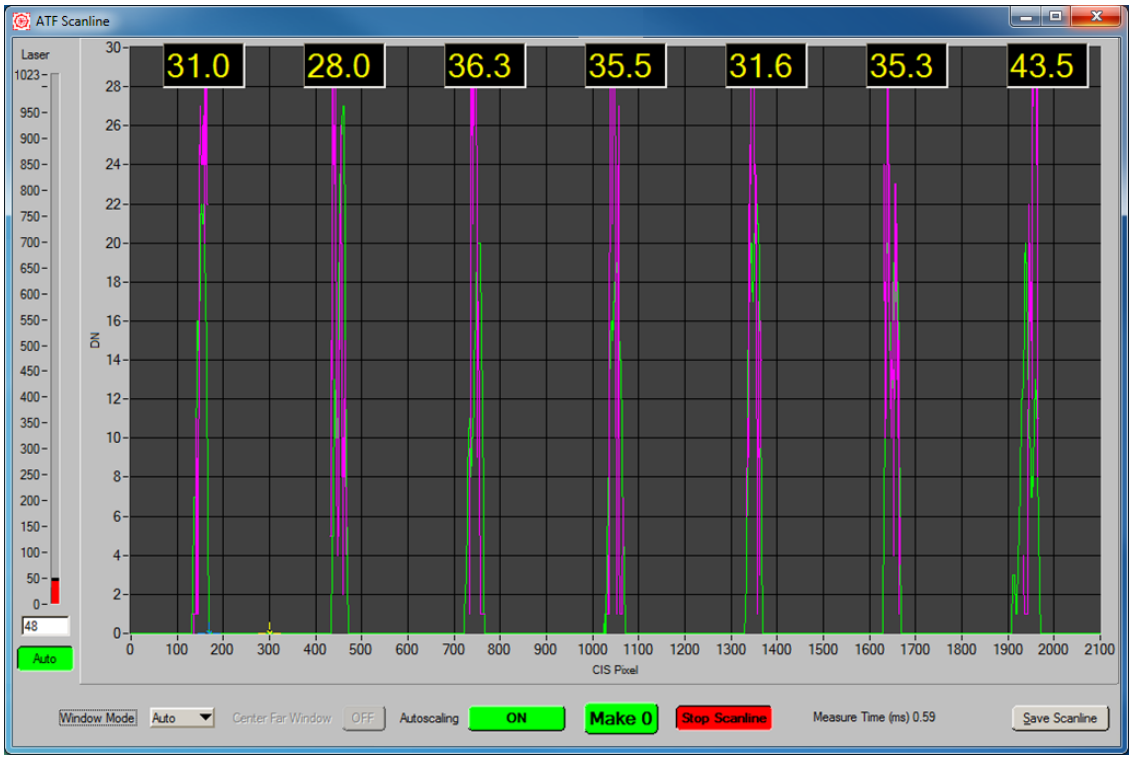


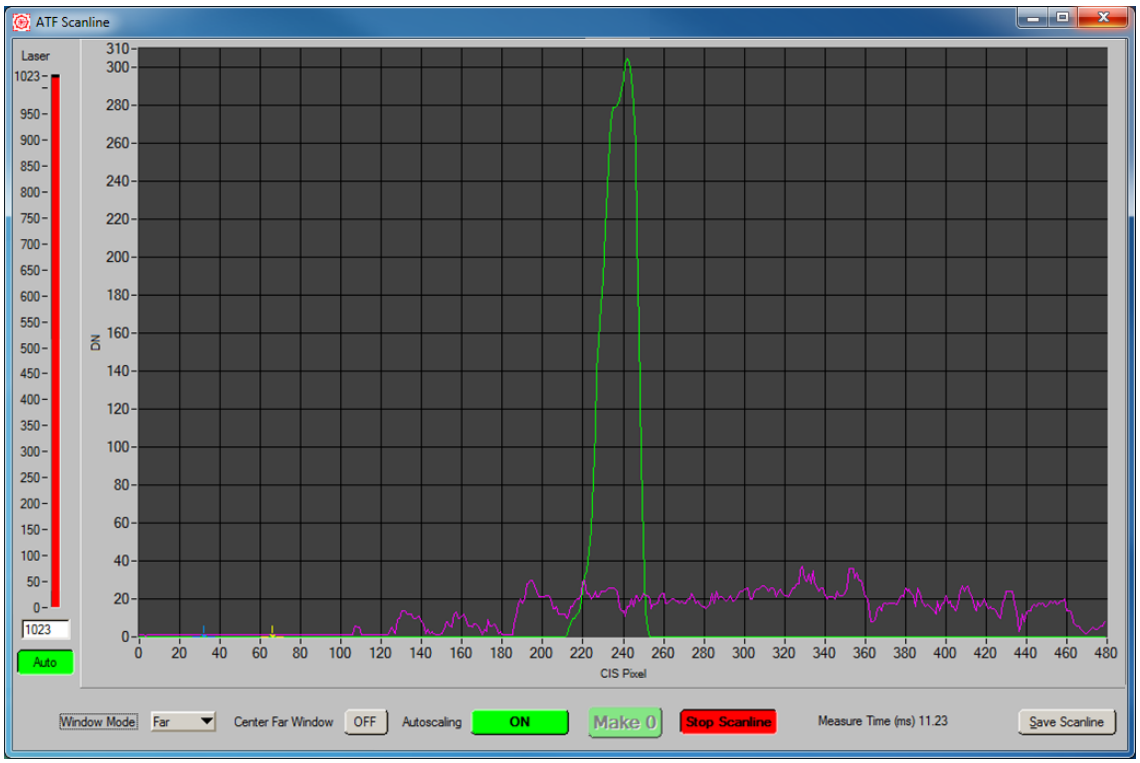
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8. Click **Make 0 CM**, which calculates and updates the **Ratio Thr.** values.
9. On the main menu, click **Move & Graph**.
10. At the bottom of the **ATF Move & Graph** window, enter the following values:
    - In the **Start (µm)** box, type **-300**.
    - In the **Step (µm)** box, type **20**.
    - In the **End (µm)** box, type **300**.
11. Click **Run**. The resulting graph should show the CF surface crossed at the 0 position, and the top glass surface visible above at a position of about 320 µm. It should also indicate a capture range of about +600 µm to the top and -300 µm to the bottom.



12. To check the peaks, on the main menu, click **Scanline**. The Scanline images should appear as shown below for **Auto** mode and **Far** mode.





Verifying CM configuration

When you are finished configuring CM using bottom validation, you can verify that it is working properly for the bottom surface as described in the following procedure.

To verify CM configuration:

1. Bring the system into focus on the bottom surface.
2. On the main panel, click **AF Off.** The system should change the laser power momentarily, jump to the top surface, and immediately return to the bottom surface. (The jump occurs only for objectives greater than 10X.)
3. Bring the system into focus on the top surface.
4. On the main panel, click **AF Off.** The system should jump back to the bottom surface.

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CHAPTER

7

Configuring video autofocus

This chapter describes how to configure the video autofocus feature.

The following topics are covered:

- *Video autofocus overview, pg. 96*
- *Setting up video autofocus, pg. 99*
- *Using video autofocus, pg. 104*

Video autofocus overview

The video autofocus feature allows the ATF 6.5 sensor to take advantage of image sensitive video techniques and precisely detect the best focal plane based on the principle of high contrast detection. Video autofocus is used specifically for high contrast surfaces, that is, patterned surfaces where there is a mixture of uniform substrate and areas that are heavily patterned, and surfaces with different heights or 3D features. Combining laser and video autofocus allows the sensor to work on virtually any surface.

Video autofocus works in two stages:

1) The system runs standard laser autofocus to bring the sample into focus based on the reflected laser signature.
2) The system turns laser autofocus off, runs video autofocus to calculate the high frequency metric content (i.e., metric), and then adjusts the Z-stage position to match the highest metric value.

The video autofocus feature is activated using the AF Video button on the ATF Console main panel (see *"Main panel overview" on page 42*). Before using the button, you must set up video autofocus for each type of sample and objective lens for which you want to use the feature as described in this chapter.

NOTE: *The video autofocus feature can bring the system into focus only when the system is stationary. Video autofocus is unable to track continuously (i.e., is not compatible with AOI Tracking) because measuring the metric value requires motion away from focus. It is also unable to track targets that move continuously in the X, Y position because the metric value can be highly dependent on the target location, which distorts the distance to focus measurement.*

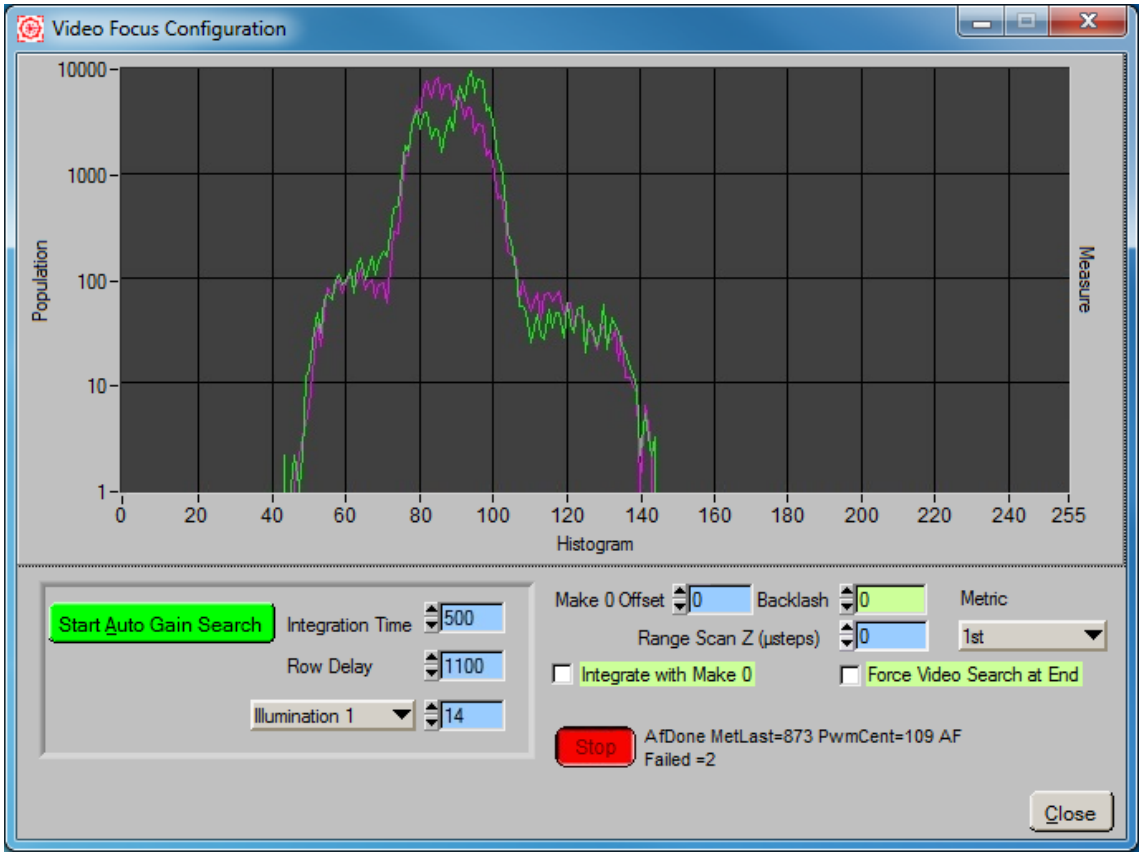
 © WDI Wise Device Inc.

Graphical modes

The plot in the Video Focus Configuration window displays graphical information about the video autofocus status.

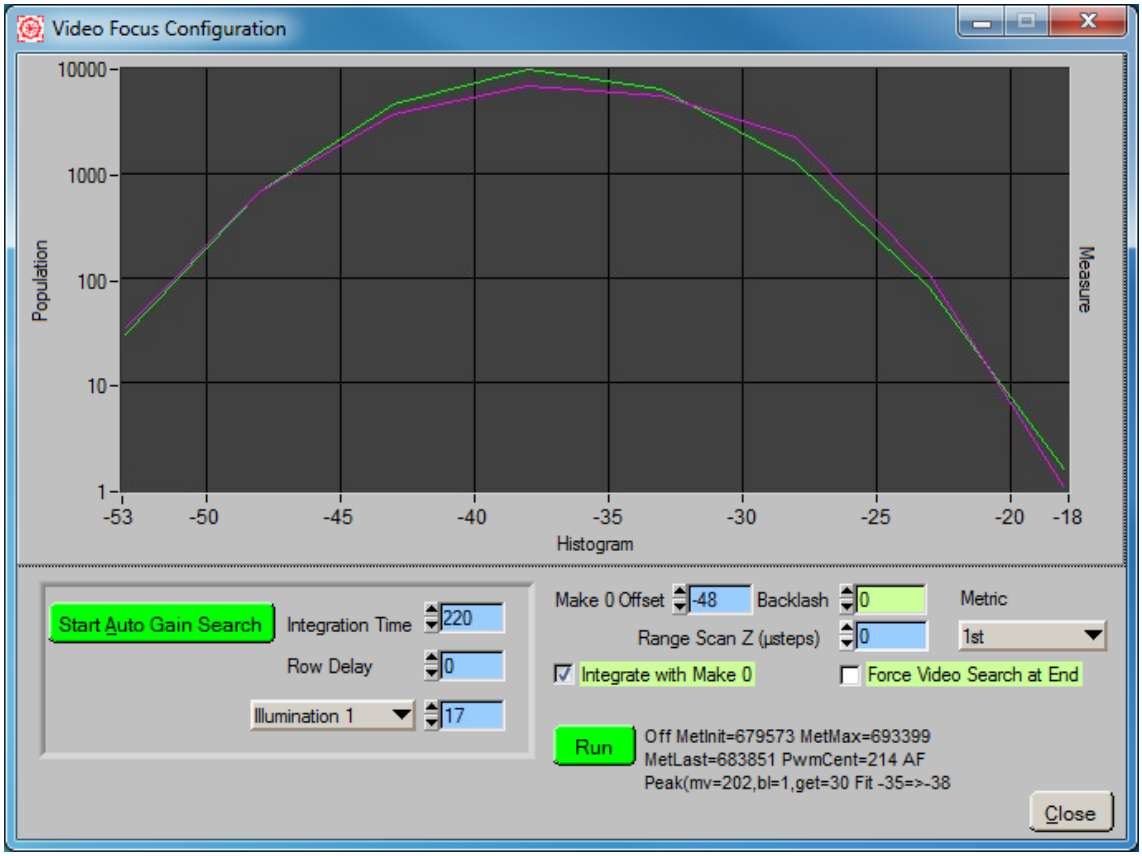
There are two graphical modes:

1) **Illumination mode** – When you click **Run** on the Video Focus Configuration window, a live video histogram shows the illumination status of the system. This mode displays the light distribution on the X-axis and the number of pixels on the Y-axis.



2) **Best fit mode** – When video autofocus is activated, the plot displays two curves:
   - Metric curve (magenta) – Shows the best focusing metric vs. absolute position.
   - Predicted curve (green) – Shows the best focusing metric prediction vs. absolute position.

   To achieve the best video focus, the Metric and Predicted curves should match as closely as possible.



Setting up video autofocus

When setting up video autofocus, the following items need to be considered:

- **Imaging condition** – The internal camera imager light sensitivity needs to be adjusted so that optimal image information can be collected. Either too much light or too little light will result in inadequate conditions and the focusing metric derived might be unavailable, noisy, or unstable.
- **Offset** – The distance calculated between the best focus position measured by the laser and the best focus position measured by the video camera's image contrast needs to be established. The smaller the offset, the better. Ideally, the offset should be 0, but in practice it is not negligible.

The imaging condition and offset are both detected automatically by the firmware during the Make 0 procedure.

NOTE: *For optimal performance, we recommend that the system setup and alignment be manually reviewed before you begin.*

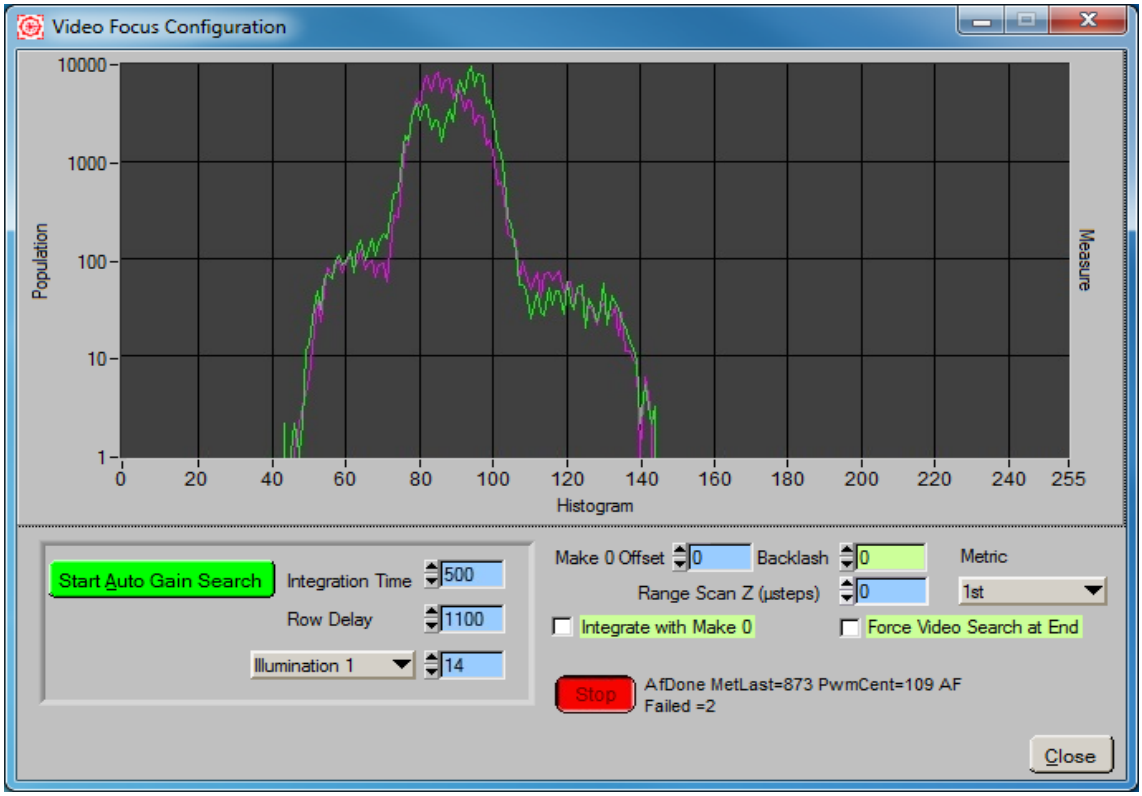
You must perform this procedure for each type of sample and each objective lens for which you want to be able to use the AF Video feature.

To set up video autofocus:

1. Start the **ATF Console** software, if it is not already started.
2. On the main panel, in the **Baud** list, select the required baud.
3. In the **Port** list, select the serial COM port where the sensor is connected.
4. In the objective lens list, select the objective lens for which you want to set up video autofocus.
5. Prepare the sample and bring it into focus using the Z-stage controls (see *"Adjusting the Z-stage position" on page 51*). Try to find a relatively smooth area of the substrate to allow the laser autofocus to perform adequately.
6. On the **Config** menu, click **Video**.

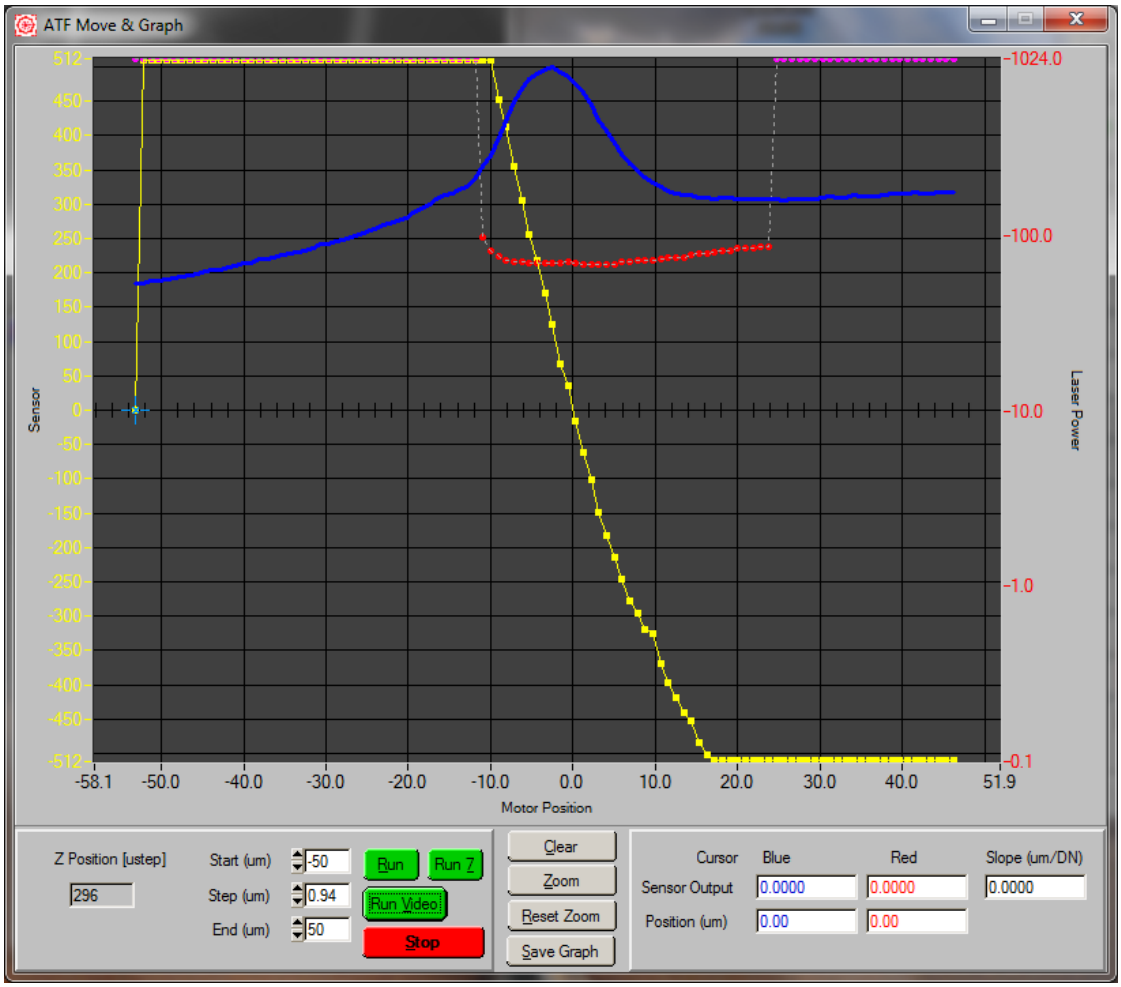
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7. In the **Video Focus Configuration** window, set the illumination settings for video autofocus:
   a. In the Illumination list, select the required illumination source, either **Illumination 1** or **Illumination 2**.
   b. Clear the **Integrate with Make 0** check box.
   c. In the **Metric** list, select the required video focus algorithm. In this case, click **1st**.
   d. Click **Start Auto Gain Search**. The system automatically determines and applies the optimal illumination gain settings (i.e., **Integration Time**, **Row Delay**, and **Illumination**) for your setup.
8. Click **Run**. This disables laser autofocus and activates Metric calculation mode. The histogram should have a pixel spread of 80 to 120 pixels. If the histogram is narrower than this, then there is insufficient light. In the box next to the Illumination source list, increase the PWM value.

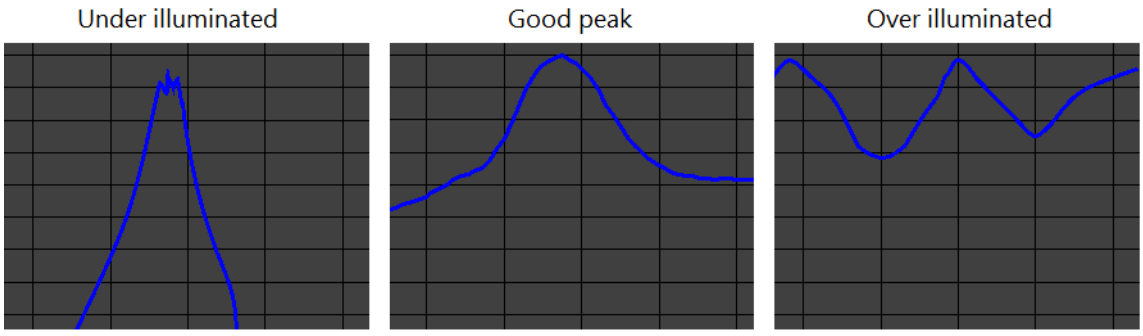


9. Click **Stop**.
10. Select the **Integrate with Make 0** check box. This will allow the system to acquire the video capture contrast around the zero position when you perform manual Make 0.
11. Perform manual Make 0. For details, see *"Performing manual Make 0" on page 53*.

12. Verify the video contrast function, as follows:
    a. On the main menu, click **Move & Graph**.
    b. In the **ATF Move & Graph** window, set the range around the focus plane, as follows:
        – In the **Start (µm)** box, type **-50**.
        – In the **End (µm)** box, type **50**.
    c. Click **Run Video**.
    d. Without changing the range, click **Run**.
    e. Verify that the blue peak (i.e., video contrast function) is centered around 0, and that the plot is unimodal and extends from -50 µm to ±20 µm.

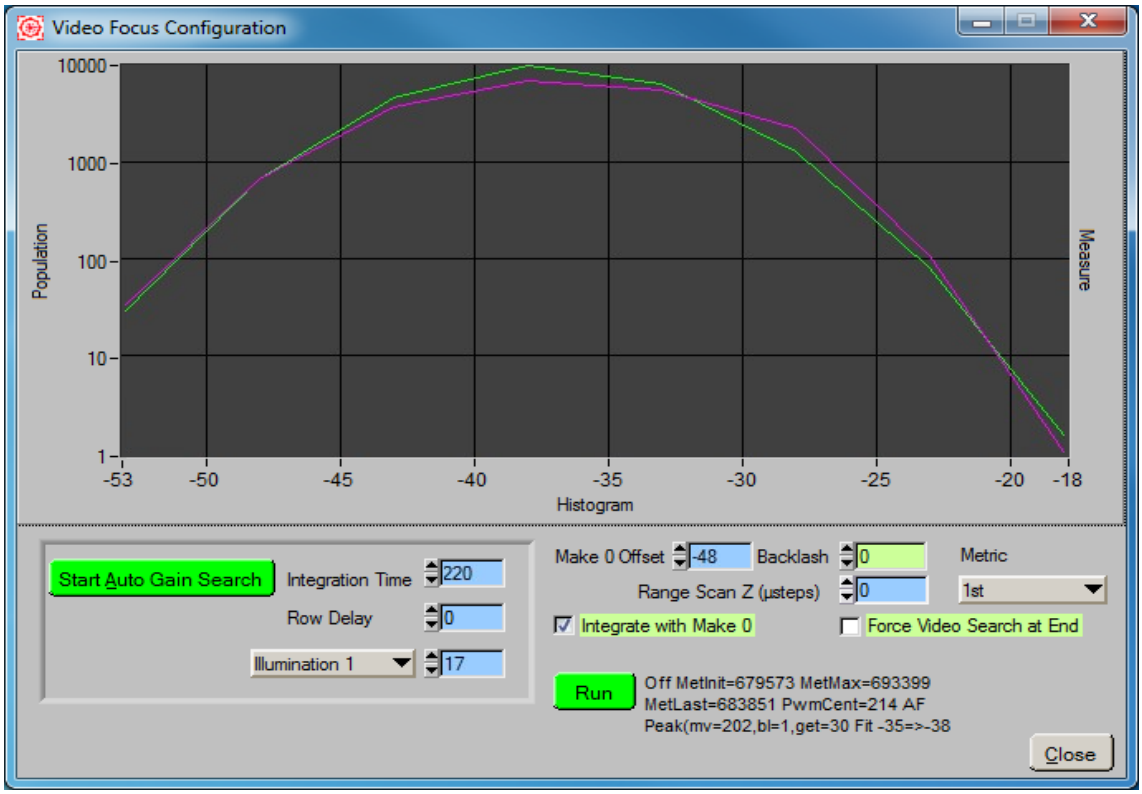


13. Observe the shape of the blue peak. It should have a single clear peak, similar to the middle image of the diagram below.
    - If the plot does not have a single clear peak, or has a very noisy signal, in the **Metric** list, select a different video focus algorithm.
    - If the tip of the peak is noisy, the system requires more illumination. In the box next to the Illumination source list, increase the PWM value.
    - If the plot rises on either side of the peak instead of dropping, the system requires less illumination. In the box next to the Illumination source list, decrease the PWM value.



14. After you have obtained a clear peak, ensure that the **Integrate with Make 0** check box is selected, and then perform manual Make 0 again.

15. In the **Video Focus Configuration** window, check the Metric (magenta) and Predicted (green) curves. The curves should be symmetrical, as shown in the following diagram.
    - If the curves are not symmetrical, try increasing the **Backlash** value.
    - If the Metric curve is too large or almost flat, try increasing the **Range Scan Z** value. This value is set to 0 by default and should be set manually only when the laser autofocus fails to reach focus.



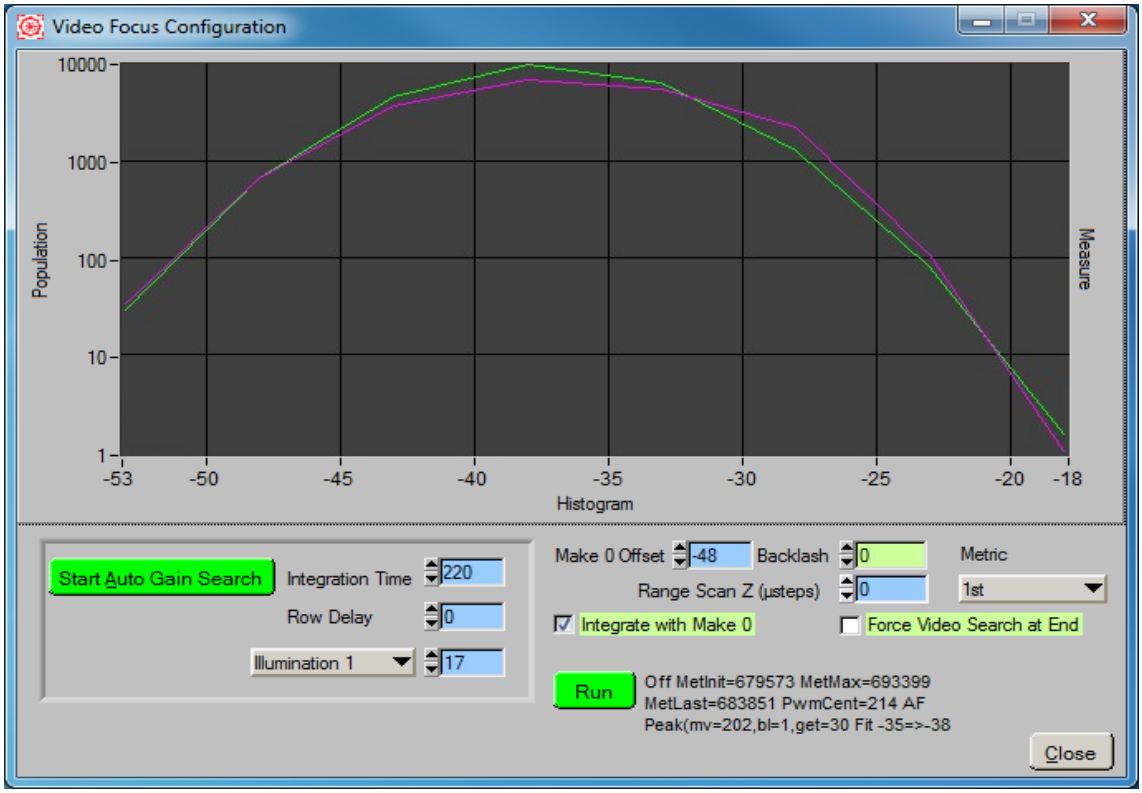
16. Perform manual Make 0 a few more times. Each time you run Make 0, ensure that the Metric (magenta) and Predicted (green) curves are symmetrical, and that the **Make 0 Offset** value is similar.
17. To verify the video autofocus performance, run AF Video at three different color pixels, or using any different samples you may have:
    a. On the main panel, click **AF Video.**
    b. On the **Config** menu, click **Video**.
    c. In the **Video Focus Configuration** window, ensure that the Metric (magenta) and Predicted (green) curves are symmetrical.

Using video autofocus

After video autofocus has been set up for the sample and objective lens that you want to use (see *"Setting up video autofocus" on page 99*), you can use it to autofocus a sample.

To use video autofocus:

1. On the main panel, click **AF Video.** The system performs video autofocus in two stages:
   a. The system runs standard laser autofocus to bring the sample into focus based on the reflected laser signature.
   b. The system turns laser autofocus off, runs video autofocus to calculate the high frequency metric content (i.e., Metric), and then adjusts the Z-stage position to match the highest Metric value.
2. On the **Config** menu, click **Video**.
3. In the **Video Focus Configuration** window, check the Metric (magenta) and Predicted (green) curves. The plot should be similar to that shown below.



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CHAPTER

8

Sensor variations

This chapter describes the variations of the ATF 6.5 sensor that are available for purchase.

The following topics are covered:

- *Sensor variation descriptions, pg. 106*
  - *ATF SA, pg. 106*
  - *ATF PZ, pg. 109*
  - *ATF with ZAA, pg. 109*
  - *ATF with Hybrid ZAA, pg. 109*
- *Filter options, pg. 109*

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Sensor variation descriptions

In addition to the full autofocus system (ATF SYS) version of the ATF 6.5 sensor, which is the main focus of this user guide, the variations described in this section are also available.

For more information regarding any of these variations, please contact our Applications Team at *applications@wdidevice.com*.

ATF SA

The standalone (SA) version of the ATF 6.5 sensor is sold without a controller and outputs an analog voltage in the range ±8 V, representing the sensor's distance-to-focus measurement. When using an ATF SA sensor, you must ensure that it is configured as AO (analog output).

If you require additional guidance on installation or use, please contact our Applications Team.

To configure the sensor as AO:

1. Ensure that the sensor is connected to a ±12 V power supply.
2. Ensure that the sensor cable is wired as shown in *Figure 22* and *Table 14*.

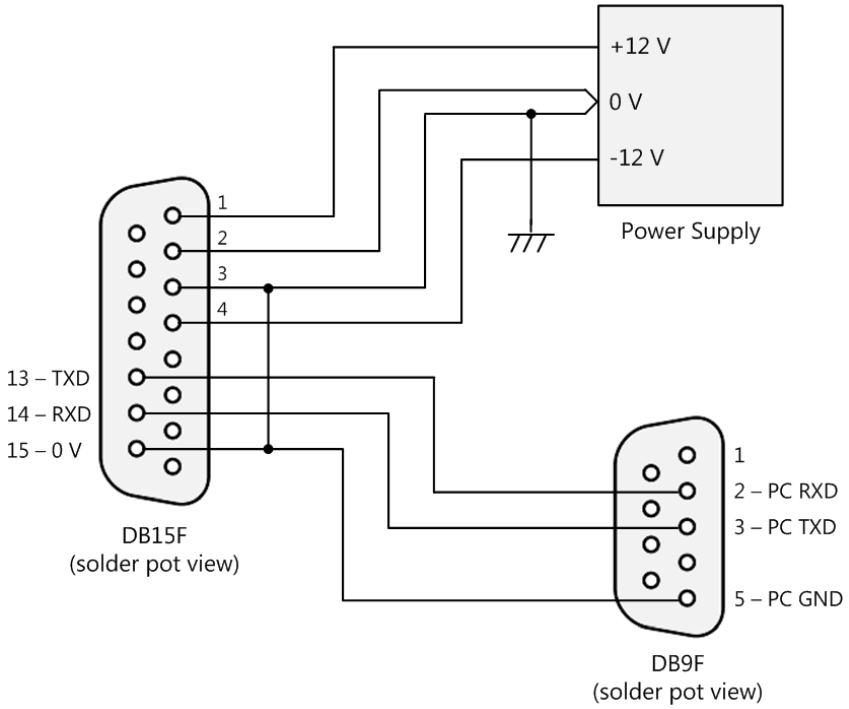


Figure 22 *ATF SA sensor cable wiring*

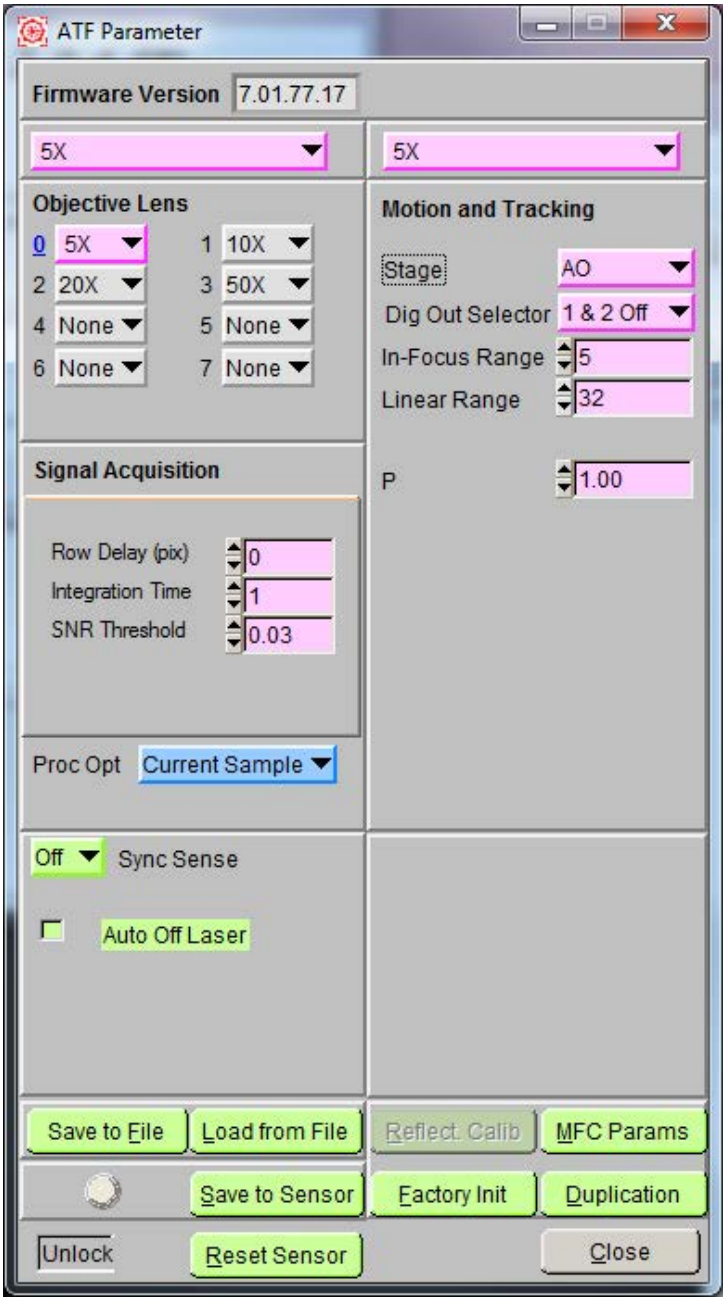
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Table 14 ATF SA sensor cable pin assignments

Pin	Signal	Function
1	+12 V	—
2	0 V	Sensor Supply Return
3	0 V	Sensor Supply Return
4	-12 V	Required for ±8 V, not required for 0-10 V
5	Analog Out	±8 V Full Range
6	0 V	Analog Return Out
7	In Capture Range	Open Collector, Digital Output
8	In Focus	—
9	Unused	Digital Input
10	Laser Enable	Digital Input
11	Camera Sync	Digital Input
12	Unused	Digital Input
13	TxD	RS-232
14	RxD	RS-232
15	0 V	RS-232 return

3. On the **Start** menu, click **ATF_Console**.
4. On the main panel, in the **Port** list, select the serial COM port where the sensor is connected.
5. In the **Baud** list, select the required baud (i.e., transmission rate).
6. On the **Config** menu, click **Parameter**.
7. At the bottom of the **ATF Parameter** dialog box, click the **Unlock** box, type **atf6**, and then press ENTER. This will unlock the parameters, including the other options on the Config menu.

8. Under **Motion and Tracking**, in the **Stage** list, select **AO**. The Parameter dialog box is updated based on your selection.



9. Repeat step 8 for each objective lens that you are using.

10. Click **Save to Sensor**.

ATF PZ

The ATF PZ is a standalone sensor with analog output (AO), similar to the ATF SA sensor, for direct connection to an external piezo Z-actuator or stage.

For information on how to set up an ATF PZ sensor, please contact Applications and request a copy of the *ATF PZ AO Setup Guide*.

ATF with ZAA

The ATF 6.5 sensor can be sold with the WDI proprietary ZAA, in which case the MCM+ is replaced with an MCZ.

For MCZ connection and usage details, refer to *UM800039 MCZ Controller User Guide*.

ATF with Hybrid ZAA

The Hybrid ZAA is designed to integrate with the ATF PZ sensor, and has both a stepper motor actuator and a piezo actuator to drive the Z-stage. The Hybrid ZAA can switch between the two types of actuation, allowing it to take advantage of both the high resolution and speed of a piezo actuator, and the longer travel range of a stepper motor actuator. The Hybrid ZAA is shipped with a specialized controller.

For information on how to install and configure the Hybrid ZAA and controller, please contact Applications and request a copy of the *Hybrid ZAA User Manual*.

Filter options

To allow ATF 6.5 sensors to perform video autofocusing effectively, no optical filters are installed within the sensor. However, for applications that have very strong illumination, a variation of the ATF 6.5 sensor (680 nm version) is available that contains a red longpass filter to help suppress the illumination entering the sensor.

For more information, please contact our Applications Team.

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